

Structure-Guided Reinforcement Learning for Partially Observed Multi-Period Portfolio Optimization with Transaction Costs

Prepared by Takahiro Kobayashi

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Takahiro Kobayashi

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Abstract

This paper develops a structure-guided reinforcement-learning framework for multi-period portfolio optimization under latent market regimes and proportional transaction costs. The central theoretical contribution is a belief-state formulation of an entropy-regularized, precommitted mean–variance control problem: under common observation covariance, Wonham filtering reduces partial observation to a fully observed belief-state diffusion, and the associated Hamilton–Jacobi–Bellman (HJB) equation identifies a Gaussian exploratory policy and a wealth-quadratic value function; this structure is then extended to regime-dependent covariance by combining an exact Gaussian Bayes correction for a discretized observation model with a moment-matched control approximation. This theoretical reduction motivates a low-dimensional actor–critic (Stage 1) in place of an unrestricted state-to-weight learner, with regime-dependent parameter estimates supplied to the online filter by a statistical jump model. Building on the frozen Stage 1 policy as a state-dependent center, Stage 2 addresses transaction costs by learning asymmetric no-trade boundaries through direct pathwise optimization of a transaction-cost-aware terminal objective; this Direct Boundary policy optimization is motivated by, but is not a numerical solution of, a transaction-cost quasi-variational-inequality (QVI) benchmark, since joint critic–intervention learning of that QVI proved numerically unstable.

The common-covariance Stage 1 and Stage 2 policies are evaluated across five independent training seeds on a common set of 1000 held-out paths. The Stage 1 policy is highly stable, with mean terminal wealth 1.1811 and a 95% Student- t training-seed interval of $[1.1808, 1.1814]$. The Direct Boundary policy achieves mean net terminal wealth 1.2009 with interval $[1.1807, 1.2211]$, exceeds daily center rebalancing by 0.0910 with interval $[0.0709, 0.1112]$, and reduces the mean trade count from 252 to 175.8 under the evaluated band-plus-leverage-cap policy. Its descriptive difference from Stage 1 net terminal wealth is 0.0198 with interval $[-0.0003, 0.0400]$; because this interval includes zero and the two stages optimize different objectives, no Stage 1-versus-Stage 2 dominance claim is made. The regime-dependent-covariance extension remains a single-seed exploratory result outside this five-seed robustness analysis. The framework is presented as a portfolio-control building block for institutional and actuarial applications rather than a complete asset–liability management model.

Keywords: continuous-time reinforcement learning; mean–variance portfolio selection; partial information; regime switching; transaction costs; no-trade region; actor–critic.

1 Introduction

Dynamic portfolio choice becomes substantially harder when three realistic features are introduced simultaneously: investment opportunities vary across persistent market regimes, the regime is not directly observed, and trading incurs transaction costs. Partial information requires the investor to infer the state of the market from prices. Multi-period objectives require decisions to account for their effect on future wealth and information. Transaction costs make frequent rebalancing suboptimal and typically replace frictionless point targets by state-dependent no-trade regions. These features lead naturally to stochastic filtering, nonlinear dynamic programming, and high-dimensional policy optimization.

The starting point of this study is the natural compatibility between portfolio choice and reinforcement learning. A portfolio rule is a policy that maps an information state into an asset allocation, while reinforcement learning is designed to learn state-dependent sequential decisions from interaction data. The central issue is therefore not only which learning algorithm to use, but also how to construct the state, policy class, and objective. Here financial domain knowledge enters all three components: filtering converts return histories into an economically interpretable regime belief; stochastic-control theory restricts the actor and critic to HJB-consistent forms; and the objective incorporates target wealth, exploration, turnover, and transaction costs. This structure is intended to improve sample efficiency, numerical stability, and interpretability relative to an unrestricted state-to-weight learner. The five-seed common-covariance experiments assess robustness to training initialization, but they do not isolate the contribution of each structural restriction through a controlled structured-versus-unstructured ablation; repeated-seed robustness and ablation are distinct open questions.

This compatibility does not imply that reinforcement learning is necessary for every low-dimensional portfolio problem. Its role here is computational: it provides a simulation-based policy-approximation method when partial information, nonlinear objectives, and transaction costs make direct dynamic programming difficult to implement.

Regime-switching portfolio control is challenging because latent-state inference, terminal credit assignment, and transaction costs interact over long horizons. The framework therefore restricts the learned policy to economically interpretable, center-relative geometry and estimates only lower-dimensional coefficients or boundaries, rather than learning an unrestricted state-to-portfolio map. For the costly problem, the frictionless solution is used as an advanced start so that Stage 2 learns a correction around an economically meaningful benchmark rather than the full policy from scratch.

Regime changes are also not limited to expected returns. Volatility levels and cross-asset correlations can change materially across market conditions, and these covariance shifts directly affect diversification, risk budgets, capital requirements, and the scale of admissible positions. For practical portfolio and risk management, covariance misspecification may therefore be at least as consequential as drift misspecification. This motivates the regime-dependent-covariance extension, while the common-covariance model is retained as the theorem-level benchmark needed to identify which HJB structures can be transferred safely.

This paper combines two complementary components. Stage 1 learns a frictionless portfolio policy under partial information. It starts from the continuous-time precommitted mean–variance formulation of Wu and Li [16], extends its structured actor–critic representation to multiple risky assets, and makes the filtering, policy, value function, and learning pipeline operational. Stage 2 first formulates the transaction-cost problem as a QVI on the enlarged state (t, x, p, w) . Because the corresponding critic-based implementation was not numerically stable, the reported method uses Center+DNNBand, a policy trained by *Direct Boundary policy optimization*: a critic-free method in which a neural policy predicts lower and upper no-trade boundaries around the frozen Stage 1 center and is trained by pathwise differentiation directly through the wealth recursion with proportional costs, rather than through a learned value function. We refer to this evaluated policy as Center+DNNBand throughout the remainder of the paper. The reported Stage 2 does not claim to solve the exact transaction-cost extension of the Stage 1 mean–variance problem: it uses the Stage 1 policy as a state-dependent structural feature and optimizes a separate transaction-cost-aware terminal objective.

The organizing principle is *structure-guided learning*. In Stage 1, the observable state is determined by filtering, the exploratory policy family is determined by the HJB, and the critic is restricted by the quadratic terminal condition. In Stage 2, classical transaction-cost theory motivates an inaction region, while the Stage 1 policy supplies a state-dependent anchor. Learning is therefore used to estimate a small set of economically meaningful objects rather than an unrestricted state-to-action map.

Across the contributions, the common methodological principle is selective function approximation: low-dimensional parametrization is used where the control equation identifies the policy form, while neural approximation is reserved for residual boundary geometry that remains analytically unresolved. The distinction between the two stages is therefore not simply parametric learning versus deep learning; it reflects how sharply the underlying control theory identifies the admissible functional form.

The paper makes the following contributions.

- (i) **A belief-state Hamilton–Jacobi–Bellman (HJB) reduction of partially observed mean–variance control, including a regime-dependent-covariance extension.** Under common observation covariance, Wonham filtering reduces the partially observed control problem to a fully observed diffusion on the belief state, and the entropy-regularized HJB equation for this belief-state problem admits a closed-form Gaussian exploratory policy together with a wealth-quadratic value function solving a lower-dimensional coefficient PDE system. We extend this reduction to regime-dependent covariance by combining an exact Gaussian Bayes correction for a discretized hidden-Markov observation model with a moment-matched, belief-averaged control approximation, deriving the resulting approximate HJB, Gaussian policy, and coefficient equations, including the additional nonlinear term that vanishes in the common-covariance benchmark. This is the central theoretical contribution of the paper; no verification or approximation-error theorem is claimed for the regime-dependent extension, and its supporting numerical experiment is a separate single-seed exploratory result outside the paper’s five-seed robustness analysis.
- (ii) **A structure-guided actor–critic implementation of the belief-state HJB reduction (Stage 1).** We make the multi-asset belief-state actor–critic operational: the Gaussian policy family, its wealth-linear mean, and the wealth-quadratic critic are fixed by the HJB reduction above, so learning is restricted to a low-dimensional set of belief/time coefficients rather than an unrestricted state-to-weight map. Regime-specific drift, covariance, and transition-intensity estimates are supplied to the online Bayesian filter by a statistical jump model applied to persistent offline labels, integrating an established regime-identification method with the HJB-guided control pipeline rather than proposing new SJM statistical theory.

- (iii) **A two-stage transaction-cost architecture with a learned no-trade boundary (Stage 2).** We formulate the costly extension of the Stage 1 problem as a quasi-variational inequality (QVI) on the enlarged state $(t, x, p, \mathbf{w})$ and derive a center-relative critic/intervention parametrization consistent with its continuation and intervention structure. Because joint QVI-residual and intervention learning was numerically unstable, the reported implementation retains the QVI’s robust structural implications—a frictionless center, an inaction region, feasibility constraints, and pathwise transaction costs—and learns asymmetric no-trade boundaries directly around the frozen Stage 1 center through Direct Boundary policy optimization: a critic-free method that optimizes a separate transaction-cost-aware terminal objective by differentiating directly through simulated wealth paths. The QVI remains the theoretical benchmark, but the evaluated boundary policy is not its numerical solution.
- (iv) **A TD-inspired implementation refinement for continuous-time actor–critic learning.** The Stage 1 actor update combines a local martingale increment, which plays a temporal-difference-like role, with a normalized full-horizon tail residual, which provides a Monte Carlo-like terminal correction. Together with the critic update, this mixed signal is designed to address finite-horizon credit assignment and provided a usable implementation in the reported experiments. It is presented as an algorithmic refinement rather than a new continuous-time RL theorem. A controlled actor-signal ablation is outside the scope of the numerical study.

The Stage 1 and Stage 2 architectures above are empirically validated across five independent training seeds, each evaluated on a common set of 1000 independently generated held-out paths. For each training seed, the same fixed evaluation scenarios are used to compute seed-level summary statistics; these are then combined across seeds into a mean, a between-seed standard deviation, and a 95% Student- t interval with four degrees of freedom. Where two policies are compared on the same paths, paired differences are computed at the evaluation-episode level before aggregation across seeds. This design quantifies sensitivity to training initialization under a fixed market-evaluation design; it is not a claim about market-distribution uncertainty. Under this protocol, the Stage 1 policy is extremely stable across training seeds, while Stage 2 shows materially greater training-seed variability. The regime-dependent-covariance extension in contribution (i) is not part of this five-seed validation and is reported separately as a single-seed exploratory result.

Actuarial relevance and potential applications. Life insurers and pension funds manage long-duration liabilities through investment strategies whose consequences accumulate over many rebalancing dates. Frequent adjustment can be impractical because of proportional transaction costs, liquidity constraints, capital limits, and governance requirements. Moreover, persistent market regimes affect not only expected returns but also volatility, correlation, credit risk, and the relative attractiveness of return-seeking and defensive assets. Partial information and trading frictions are therefore central features of the institutional investment problem rather than secondary implementation details.

Within the present framework, the terminal target z can be interpreted as a reduced-form accumulation, reserve, or funding objective, including objectives linked to credited-rate commitments. The posterior belief p_t provides a probabilistic summary of the prevailing market environment, the Stage 1 policy specifies a regime-aware frictionless risk exposure, and the Stage 2 no-trade region translates the trade-off between tracking that exposure and limiting costly turnover into a transparent rebalancing rule that can support governance oversight. In a fixed-income or liability-driven-investment application, the same architecture could support regime-aware allocation between return-seeking credit and defensive government-bond sleeves, or govern the rebalancing of duration and credit overlays. These examples are prospective applications of the architecture; they are not evaluated in the numerical experiments reported here.

The model is not a complete asset–liability management system. It does not include stochastic liability cash flows, interest-rate-sensitive liability valuation, solvency capital, accounting constraints, taxes, cash-flow matching, or endogenous market impact. A broader ALM formulation could augment the state with liability values, funding ratios, and yield-curve factors; replace the terminal-wealth criterion with a surplus- or liability-relative objective; and impose institution-specific capital, liquidity, and asset-allocation constraints. The present contribution should therefore be viewed as a regime-aware portfolio-control building block for such a system.

Table 1 summarizes how the main components are positioned relative to prior literature. The table supplements, rather than replaces, the contribution statements above.

Table 1: Positioning of the paper relative to prior literature.

Component	Prior literature	This paper
Stage 1 control	One-risky-asset hidden-regime benchmark; common-covariance multi-asset extension	Operational multi-asset actor-critic with explicit coefficient PDEs and implementation details
Regime estimation	SJM-based persistent regime identification	SJM labels integrated into parameter estimation for online belief-state control
Covariance structure	Common observation covariance in the theorem-level benchmark	Discretized Markov prediction and exact Gaussian Bayes correction plus an approximate belief-averaged HJB control layer
Transaction costs	Classical no-trade regions and HJB-initialized boundary networks	Belief-conditioned two-stage learning with a frozen Stage 1 center and Center+DNNBand
Actor update	Martingale-based continuous-time policy evaluation and gradient methods	Mixed local martingale and full-horizon rollout signal for finite-sample learning

2 Related Literature

2.1 Continuous-time reinforcement learning and mean-variance control

The continuous-time reinforcement-learning foundation used here follows the exploratory stochastic-control formulation of Wang, Zariphopoulou, and Zhou [15]. Randomized controls and entropy regularization convert exploration into part of the control objective. In linear-quadratic problems, this construction naturally produces Gaussian policies. Wang and Zhou [14] apply the framework to continuous-time mean-variance portfolio selection and show how analytical structure can substantially reduce the number of learned parameters.

Jia and Zhou [4] develop a martingale characterization of continuous-time policy evaluation. A central insight is that a naive mean-square temporal-difference error converges toward quadratic variation and is not, by itself, the appropriate continuous-time value-fitting objective. They instead derive martingale-loss and martingale-orthogonality formulations that recover continuous-time analogues of Monte Carlo and TD methods. Their companion paper [5] derives policy-gradient representations and offline and online actor-critic algorithms. The policy-gradient problem is recast through policy evaluation, while online updates use martingale orthogonality and stochastic approximation. These results motivate the martingale-based critic and actor signals used in Stage 1.

2.2 Partial information and regime-switching portfolios

Partial-information portfolio problems separate state estimation from control when an appropriate filtering reduction is available. Wonham [17] derives the finite-state nonlinear filter used in the common-covariance benchmark, while Krishnamurthy [7] develops the broader POMDP perspective in which the posterior distribution becomes the fully observed belief state. Xiong and Zhou [18] establish a separation principle for continuous-time mean-variance selection under partial information, and Elliott, Siu, and Badescu [3] study estimation and filtering under hidden Markov regimes.

For observable market regimes, Zhou and Yin [19] derive continuous-time mean-variance results and efficient-frontier implications. Under unobservable Markov-modulated drift, Rieder and Bäuerle [10] study portfolio optimization after filtering and identify the interaction between current opportunities and hedging against belief changes. These papers clarify the role of regime information and motivate the myopic-hedging decomposition used in the present Stage 1 actor.

The closest benchmark is Wu and Li [16]. They formulate an entropy-regularized, precommitted mean-variance problem with an unobservable market regime, use Wonham filtering to obtain a belief-state problem, and derive semi-analytical representations of the value function and Gaussian optimal policy. The unknown coefficient functions solve lower-dimensional parabolic PDEs and are approximated by polynomials inside an actor-critic algorithm. The present Stage 1 follows this logic, makes the multi-asset extension explicit, and adds a practical regime-estimation pipeline and a regime-dependent covariance experiment.

2.3 Statistical jump models for regime identification

Statistical jump models estimate persistent market regimes by clustering temporally constructed features while directly penalizing changes in the latent label sequence. This formulation offers a non-likelihood-based alternative to a conventional hidden Markov model and is particularly attractive when financial regimes are persistent, imbalanced, or supported by limited data. Aydinhan, Kolm, Mulvey, and Shu [1] review the discrete statistical jump model and extend its discrete hidden state to a probability vector on the regime simplex. Their continuous statistical jump model introduces smooth regime probabilities, derives a probabilistic interpretation of the transition penalty, and adds a mode-loss penalty that encourages decisive regime assignments. Their simulations and market experiments report improved robustness relative to discrete jump models and standard HMM estimation, especially under regime imbalance and short samples.

The present paper uses the simpler discrete two-state SJM rather than the continuous extension of Aydinhan et al. Its role is also narrower: the SJM provides a persistent offline segmentation from which regime-specific drifts, covariances, and empirical transition rates are estimated. It does not replace the online Wonham or Gaussian Bayes filter used to construct the belief state for control. Thus, the contribution here is the integration of SJM-based estimation with an HJB-guided portfolio-learning architecture, not a new SJM estimator or a new statistical consistency result.

The choice of SJM does not imply that it uniformly dominates hidden Markov models. HMMs provide a coherent probabilistic observation model and naturally generate filtered and smoothed regime probabilities. The discrete SJM used here instead offers direct control over label persistence through the jump penalty and produces regime prototypes in an economically constructed feature space. This makes the persistence penalty, feature centroids, and resulting offline segmentation particularly transparent when the objective is to identify persistent regimes rather than to estimate a complete generative return model.

The two approaches play complementary roles in the present pipeline. SJM is used offline to segment historical or simulated observations and estimate regime-dependent moments and transition intensities. Online allocation does not treat the resulting hard SJM label as an observed state; it uses a Bayesian filter to update the posterior regime probability from newly observed returns. This separation avoids supplying an ex-post label directly to the controller and preserves the partial-information structure of the investment problem.

2.4 Transaction costs and no-trade regions

The no-trade-region principle originates in classical proportional-cost portfolio theory. Magill and Constantinides [8] show that investors optimally tolerate deviations from frictionless target proportions. Davis and Norman [2] provide a rigorous infinite-horizon formulation and numerical characterization, and Shreve and Soner [11] establish existence, uniqueness, and regularity results using viscosity-solution methods. Taksar, Klass, and Assaf [13] provide a diffusion model with brokerage fees in which the risky-to-safe wealth ratio is kept inside an interval with the minimum intervention needed at its boundaries. Together, these studies supply the economic and QVI intuition for Stage 2: proportional costs create an inaction region rather than a continuously tracked point target.

High-dimensional PDE methods motivate neural approximations when grid-based QVI solvers become impractical. Han, Jentzen, and E [6] and Sirignano and Spiliopoulos [12] develop deep-learning methods for high-dimensional PDEs. In portfolio optimization, Mulvey et al. [9] solve a frictionless HJB system for mean-reverting Ornstein–Uhlenbeck assets to obtain an advanced start and then use a feedforward neural network to approximate state- and time-dependent no-trade boundaries. Their experiments demonstrate scalability to high-dimensional collections of correlated pair trades. The present work adopts the same broad principle—combine a tractable frictionless solution with a learned transaction-cost correction—but differs in three respects. The underlying market is regime-switching rather than OU, the state contains a filtered belief, and the reported Stage 2 network directly optimizes a separate terminal utility and regularization objective through simulated wealth paths. Accordingly, the Stage 1 center is a structural feature rather than the exact zero-cost optimizer of the Stage 2 objective.

3 Decision Framework and Partially Observed Market

3.1 From partial observation to belief-state control

A reinforcement-learning problem is described by an information state, an action, a policy, and an objective. In this portfolio application, the action is an allocation of risky exposure, while the information state must summarize what is relevant for both future wealth dynamics and future decisions. In Stage 1 the observable

control state is (t, X_t, p_t) , where X_t is discounted wealth and p_t is the posterior probability of regime 1. In Stage 2 the current holdings or portfolio weights are added because transaction costs depend on the distance between pre-trade and post-trade positions.

The underlying regime I_t is not observed directly. The investor observes the price or return history $\mathcal{F}_t^S$, so the original problem is a partially observed Markov decision process. Under the filtering assumptions used below, the posterior distribution of the finite-state regime is a sufficient statistic for that observation history. The problem can therefore be reformulated as a fully observed Markov control problem on the *belief state* [7]. This transformation does not eliminate regime uncertainty: it represents that uncertainty explicitly through

$$p_t = \mathbb{P}(I_t = 1 \mid \mathcal{F}_t^S). \quad (1)$$

The belief is thus an information state for control, not merely a regime-classification score.

The actor specifies the portfolio decision conditional on the information state. The critic approximates the conditional expected continuation objective from that state, and actor-critic learning uses the critic signal to improve the actor. Reinforcement learning is employed here as a simulation-based numerical approximation method; stochastic filtering and control theory determine the state representation and restrict the functional forms being learned.

Table 2: Reinforcement-learning objects in the portfolio-control problem.

Object	Stage 1: frictionless control	Stage 2: costly rebalancing
Information state	(t, X_t, p_t)	$(t, X_t, p_t, \mathbf{w}_t)$
Action	Risky-dollar position $\mathbf{u}_t$	Post-trade weight or intervention target
Actor / policy	Gaussian distribution conditional on the belief state	Belief- and horizon-dependent no-trade boundaries
Critic / value model	Expected exploratory mean-variance cost-to-go	QVI value function in the theoretical design; no separate critic in the reported Direct Boundary implementation
Learning objective	Martingale and terminal cost-to-go signals	Differentiable terminal utility plus turnover, gap, and leverage regularization
Learning role	Estimate low-dimensional policy and value coefficients	Estimate boundary geometry around the frozen Stage 1 center

Figure 1 summarizes the information flow. The hidden regime affects observed returns but is not supplied directly to the controller. The filter compresses the observation history into p_t , which enters the state used by the actor or boundary policy.

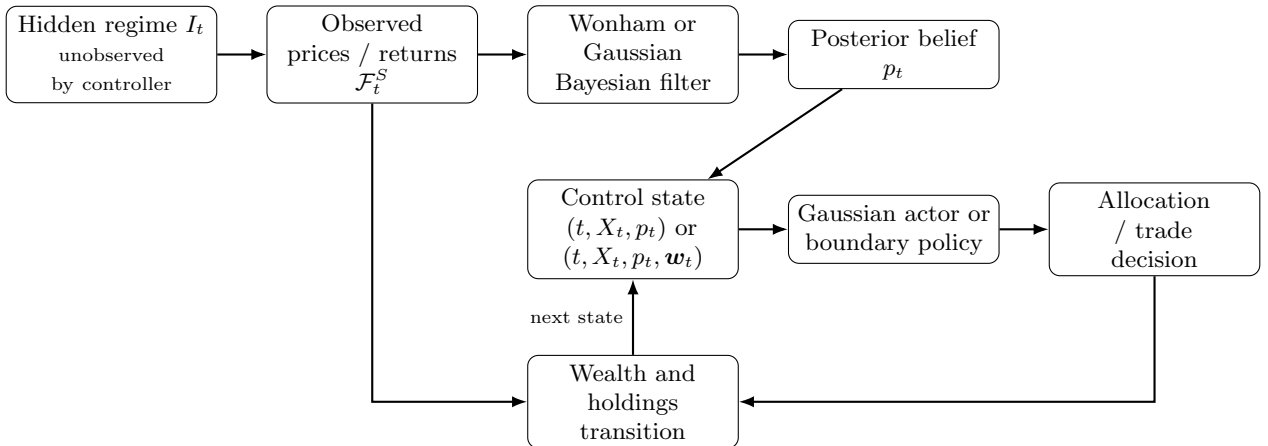


Figure 1: Belief-state construction and portfolio-control information flow.

3.2 Selective use of deep function approximation

Neural networks provide flexible approximations to high-dimensional nonlinear functions, but flexibility is not costless: a larger function class can increase optimization variance, data requirements, computational burden, and the difficulty of understanding the resulting decision rule. The present framework therefore does not use deep learning uniformly across the two stages. Model complexity is allocated according to the extent to which

stochastic-control theory identifies the relevant functional form. Deep learning is not treated as inherently uninterpretable; interpretability depends substantially on what the network is permitted to represent.

In Stage 1, the HJB implies a Gaussian exploratory policy, a policy mean that is linear in wealth relative to the multiplier, and a value function that is quadratic in wealth. A low-dimensional parametrization is therefore used instead of a generic deep state-to-weight network. This choice preserves the analytically identified structure and avoids asking simulation data to relearn a functional form already supplied by the control problem. The principal motivations are identifiability, sample efficiency, and numerical discipline; reduced computation and improved interpretability are additional benefits rather than the primary justification.

In Stage 2, the transaction-cost QVI identifies the enlarged state, continuation-versus-intervention logic, proportional-cost wealth transition, and qualitative role of an inaction region around a frictionless center. It does not characterize the finite-horizon multidimensional lower and upper boundaries sufficiently sharply for the present implementation. The reported neural network is therefore not asked to learn an unrestricted portfolio policy. It learns only the residual time-, belief-, wealth-, and holdings-dependence of the boundaries, subject to center-relative geometry, ordering, and feasibility constraints. The architecture is thus structurally parametric in its economic geometry but flexible in the remaining state dependence.

3.3 Regime-switching market

The latent chain is a parsimonious representation of persistent market conditions, such as favorable versus stressed, risk-on versus risk-off, or normal versus high-volatility environments. It is not a literal claim that financial markets have only two economic states. A continuous-time Markov chain captures random switching while preserving tractability: λ_i is the transition intensity out of regime i , and the expected holding time in that regime is $1/\lambda_i$. The two-state specification is adopted for interpretation and computational economy; additional states, duration dependence, or non-Markov persistence are possible extensions.

The economic labels are determined by the estimated drift and covariance parameters rather than imposed in advance. In the common-covariance benchmark only expected returns change by regime. The later extension also allows volatility and cross-asset correlation to change, reflecting their direct role in diversification and risk concentration.

Let $I_t \in \{1, 2\}$ be a latent continuous-time Markov chain with generator

$$Q = \begin{pmatrix} -\lambda_1 & \lambda_1 \\ \lambda_2 & -\lambda_2 \end{pmatrix}, \quad \lambda_1, \lambda_2 > 0. \quad (2)$$

There are d risky assets and one risk-free asset with rate r . In the common-covariance benchmark,

$$d\mathbf{S}_t = \text{Diag}(\mathbf{S}_t)(\boldsymbol{\mu}_t dt + \sigma d\mathbf{B}_t), \quad \boldsymbol{\mu}_t = \boldsymbol{\mu}^{(1)}\mathbf{1}_{\{I_t=1\}} + \boldsymbol{\mu}^{(2)}\mathbf{1}_{\{I_t=2\}}, \quad (3)$$

where $\sigma \in \mathbb{R}^{d \times d}$ is nonsingular and $\boldsymbol{\Sigma} = \sigma\sigma^\top$. Define

$$\boldsymbol{\rho}^{(i)} = \sigma^{-1}(\boldsymbol{\mu}^{(i)} - r\mathbf{1}), \quad i = 1, 2. \quad (4)$$

If $\mathbf{u}_t \in \mathbb{R}^d$ denotes discounted risky-dollar positions, discounted wealth under full information satisfies

$$dX_t^u = \mathbf{u}_t^\top \sigma(\boldsymbol{\rho}_t dt + d\mathbf{B}_t). \quad (5)$$

The regime is hidden. The investor observes prices and uses the posterior belief

$$p_t = \mathbb{P}(I_t = 1 \mid \mathcal{F}_t^S). \quad (6)$$

The original control problem is therefore a partially observed Markov decision problem. Under the common-covariance assumption, the posterior p_t is a sufficient statistic for the observation history, so the POMDP is converted into a fully observed control problem on the belief state.

Proposition 1 (Belief-state reduction under common covariance). *Suppose the regime follows (2), the asset model is (3), σ is nonsingular and common across regimes, and controls are adapted to the price filtration. Then the innovation process in (8) is Brownian with respect to that filtration, the posterior belief satisfies (11), and discounted wealth satisfies (10). Consequently, (t, X_t, p_t) is a fully observed Markov information state for Stage 1.*

Proof sketch. Define the filtered drift and the innovation process by

$$\hat{\boldsymbol{\mu}}(p) = p\boldsymbol{\mu}^{(1)} + (1-p)\boldsymbol{\mu}^{(2)}, \quad (7)$$

$$d\hat{\mathbf{B}}_t = \sigma^{-1} d \log \mathbf{S}_t - \sigma^{-1} (\hat{\boldsymbol{\mu}}(p_t) - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma})) dt. \quad (8)$$

Because the observation covariance is the same in both regimes, the innovations theorem implies that $\hat{\mathbf{B}}$ is Brownian in the price filtration [17]. Indeed,

$$d \log \mathbf{S}_t = (\boldsymbol{\mu}_t - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma})) dt + \sigma d\mathbf{B}_t = (\hat{\boldsymbol{\mu}}(p_t) - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma})) dt + \sigma d\hat{\mathbf{B}}_t. \quad (9)$$

Substituting this representation into discounted wealth and applying the two-state Wonham filter gives

$$dX_t^u = \mathbf{u}_t^\top \sigma (\hat{\boldsymbol{\rho}}(p_t) dt + d\hat{\mathbf{B}}_t), \quad (10)$$

$$dp_t = a(p_t) dt + b(p_t)^\top d\hat{\mathbf{B}}_t, \quad (11)$$

where

$$\hat{\boldsymbol{\rho}}(p) = p\boldsymbol{\rho}^{(1)} + (1-p)\boldsymbol{\rho}^{(2)}, \quad a(p) = \lambda_2 - (\lambda_1 + \lambda_2)p, \quad b(p) = p(1-p)(\boldsymbol{\rho}^{(1)} - \boldsymbol{\rho}^{(2)}). \quad (12)$$

Thus (X_t, p_t) is the observable state for Stage 1. The shared innovation in (10)–(11) creates a wealth–belief cross-variation term in the HJB. $\square$

For simulation with step Δt , the filter is implemented by a prediction–correction recursion. The Markov prediction is

$$p_{k|k-1} = p_{k-1}(1 - \lambda_1 \Delta t) + (1 - p_{k-1})\lambda_2 \Delta t. \quad (13)$$

Under the common-covariance observation model, the log-return likelihood is

$$y_k | I_k = i \sim \mathcal{N} \left(\left(\boldsymbol{\mu}^{(i)} - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma}) \right) \Delta t, \boldsymbol{\Sigma} \Delta t \right), \quad (14)$$

and the two predicted log weights are normalized with a log-sum-exp calculation. The controller never observes I_k ; it receives (t_k, X_k, p_k) .

3.4 Exploratory mean–variance objective

For a target expected terminal wealth z , the precommitted mean–variance problem is

$$\min_{\mathbf{u}} \text{Var}(X_T^u) \quad \text{subject to} \quad \mathbb{E}[X_T^u] = z. \quad (15)$$

Here “precommitted” means that the policy is selected at the initial date for the terminal mean–variance criterion and is not subsequently reoptimized as a time-consistent equilibrium strategy. Introducing a Lagrange multiplier ω gives the saddle formulation

$$\sup_{\omega \in \mathbb{R}} \inf_{\mathbf{u}} \{ \mathbb{E}[(X_T^u - \omega)^2] - (\omega - z)^2 \}. \quad (16)$$

For a fixed multiplier ω , the inner control problem is the entropy-regularized problem developed below. The multiplier is subsequently updated by dual ascent so that the terminal-wealth mean constraint is satisfied; this outer step is given explicitly in (55). Let $\boldsymbol{\pi}(\cdot | t, x, p)$ be a stochastic policy density and let $\mathcal{H}(\boldsymbol{\pi}) = - \int \boldsymbol{\pi}(\mathbf{u}) \ln \boldsymbol{\pi}(\mathbf{u}) d\mathbf{u}$ be its differential entropy. The exploratory objective is

$$V^\pi(t, x, p) = \mathbb{E}_{t,x,p} \left[(X_T^\pi - \omega)^2 - \Lambda \int_t^T \mathcal{H}(\boldsymbol{\pi}_s) ds \right] - (\omega - z)^2, \quad (17)$$

and the optimal value is $V^* = \inf_{\boldsymbol{\pi}} V^\pi$. The entropy coefficient $\Lambda > 0$ controls exploration.

4 Stage 1: HJB-Guided Actor–Critic Learning

4.1 Exploratory HJB and Gaussian policy

Assume that σ is nonsingular, admissible policies have finite second moments and finite differential entropy, and the candidate value satisfies the regularity required for Itô’s formula. For a fixed deterministic action $\mathbf{u}$, the observable state dynamics imply

$$d\langle X \rangle_t = \mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{u} dt, \quad (18)$$

$$d\langle p \rangle_t = \|b(p)\|^2 dt, \quad (19)$$

$$d\langle X, p \rangle_t = \mathbf{u}^\top \sigma b(p) dt. \quad (20)$$

Consequently, the controlled generator applied to a smooth function v is

$$\mathcal{L}^{\mathbf{u}} v = v_t + a(p)v_p + \frac{1}{2}\|b(p)\|^2 v_{pp} + \mathbf{u}^\top \sigma \hat{\rho}(p)v_x \quad (21)$$

$$+ \frac{1}{2}\mathbf{u}^\top \Sigma \mathbf{u} v_{xx} + \mathbf{u}^\top \sigma b(p)v_{xp}. \quad (22)$$

The last term is essential: it is generated by the common innovation driving both wealth and belief. Averaging the action-dependent terms in (22) under a policy density gives the control-dependent part of the exploratory HJB,

$$\min_{\boldsymbol{\pi}} \int_{\mathbb{R}^d} \left[\frac{1}{2}\mathbf{u}^\top \Sigma \mathbf{u} v_{xx} + \mathbf{u}^\top \sigma \hat{\rho}(p)v_x + \mathbf{u}^\top \sigma b(p)v_{xp} + \Lambda \ln \pi(\mathbf{u}) \right] \pi(\mathbf{u}) d\mathbf{u}. \quad (23)$$

The remaining terms are $v_t + a(p)v_p + \frac{1}{2}\|b(p)\|^2 v_{pp}$, and the terminal condition is

$$v(T, x, p) = (x - \omega)^2 - (\omega - z)^2. \quad (24)$$

To solve the minimization over densities, define

$$A = v_{xx} \Sigma, \quad q = \sigma [\hat{\rho}(p)v_x + b(p)v_{xp}]. \quad (25)$$

If $v_{xx} > 0$, then A is positive definite. Let

$$m = -A^{-1}q, \quad \varphi^*(\mathbf{u}) = \mathcal{N}(\mathbf{u} \mid m, \Lambda A^{-1}). \quad (26)$$

Completing the square and using the Gibbs variational identity gives

$$\begin{aligned} & \int_{\mathbb{R}^d} \left[\frac{1}{2}\mathbf{u}^\top A \mathbf{u} + q^\top \mathbf{u} + \Lambda \log \pi(\mathbf{u}) \right] \pi(\mathbf{u}) d\mathbf{u} \\ &= \Lambda \text{KL}(\pi \parallel \varphi^*) - \frac{1}{2}q^\top A^{-1}q - \frac{\Lambda}{2} \log \det(2\pi \Lambda A^{-1}). \end{aligned} \quad (27)$$

The Gibbs identity yields the following pointwise characterization.

Proposition 2 (Gaussian exploratory Hamiltonian optimizer). *If $v_{xx} > 0$ and the policy density has finite second moment and finite differential entropy, the unique minimizer of the entropy-regularized Hamiltonian (23) is*

$$\pi^*(\cdot \mid t, x, p) = \mathcal{N}\left(-\frac{(\sigma^\top)^{-1}[\hat{\rho}(p)v_x + b(p)v_{xp}]}{v_{xx}}, \frac{\Lambda}{v_{xx}} \Sigma^{-1}\right). \quad (28)$$

Proof. Equation (27) writes the objective as a constant plus $\Lambda \text{KL}(\pi \parallel \varphi^*)$. The Kullback–Leibler divergence is nonnegative and vanishes only when $\pi = \varphi^*$ almost everywhere, which gives (28). $\square$

This proposition is the main reduction used by the actor: the policy family is Gaussian by construction, and learning focuses on its low-dimensional mean and covariance coefficients.

Completing the square also gives the optimized HJB explicitly:

$$\begin{aligned} 0 &= v_t + a(p)v_p + \frac{1}{2}\|b(p)\|^2 v_{pp} - \frac{\|\hat{\rho}(p)v_x + b(p)v_{xp}\|^2}{2v_{xx}} \\ &\quad - \frac{\Lambda}{2} \log \det\left(\frac{2\pi\Lambda}{v_{xx}} \Sigma^{-1}\right). \end{aligned} \quad (29)$$

The first negative term is the minimized quadratic Hamiltonian and the logarithmic term is the entropy contribution evaluated at the Gaussian density.

4.2 Quadratic value representation

The terminal condition motivates

$$v(t, x, p) = \frac{(x - \omega)^2}{f(t, p)} + g(t, p) - (\omega - z)^2, \quad f(t, p) > 0. \quad (30)$$

Then

$$v_x = \frac{2(x - \omega)}{f}, \quad v_{xx} = \frac{2}{f}, \quad v_{xp} = -2(x - \omega) \frac{f_p}{f^2}. \quad (31)$$

The remaining derivatives needed for coefficient matching are

$$v_t = -(x - \omega)^2 \frac{f_t}{f^2} + g_t, \quad (32)$$

$$v_p = -(x - \omega)^2 \frac{f_p}{f^2} + g_p, \quad (33)$$

$$v_{pp} = -(x - \omega)^2 \frac{f_{pp}}{f^2} + 2(x - \omega)^2 \frac{f_p^2}{f^3} + g_{pp}. \quad (34)$$

Substitution into (28) yields

$$\pi^*(\cdot \mid t, x, p) = \mathcal{N} \left(-(x - \omega)(\sigma^\top)^{-1} \left[\hat{\rho}(p) - b(p) \frac{f_p(t, p)}{f(t, p)} \right], \frac{\Lambda f(t, p)}{2} \Sigma^{-1} \right). \quad (35)$$

The policy mean separates into a myopic demand and an intertemporal hedging demand:

$$\begin{aligned} \mathbb{E}[\mathbf{u}_t \mid t, x, p] = & -(x - \omega)(\sigma^\top)^{-1} \hat{\rho}(p) \\ & + (x - \omega)(\sigma^\top)^{-1} b(p) \frac{f_p(t, p)}{f(t, p)}. \end{aligned} \quad (36)$$

The first component responds to the current filtered Sharpe ratio, whereas the second hedges changes in future investment opportunities induced by belief fluctuations. The exploration covariance is proportional to $\Lambda f(t, p)$ and inversely proportional to the return covariance. As $\Lambda \downarrow 0$, the Gaussian policy converges weakly to a point mass at its mean, recovering the classical feedback control under the same regularity conditions.

The Gaussian family is also supported by policy improvement. Given the value V^π of an admissible policy, replacing that policy pointwise by the Gaussian obtained from (28) with $v = V^\pi$ minimizes the local exploratory Hamiltonian. If the updated policy remains admissible, the standard verification argument gives $V^{\pi'} \leq V^\pi$ under the cost-minimization convention. Thus restricting learning to structured Gaussian actors is not merely a convenient distributional assumption.

The first two moments of this distribution are respectively linear and quadratic in $(x - \omega)$, so the HJB remains closed within the wealth-quadratic class. The original PDE in (t, x, p) therefore reduces to coefficient equations in (t, p) .

Proposition 3 (Wealth-quadratic closure). *Under the assumptions of Proposition 2, suppose a sufficiently regular solution of the optimized HJB has the form (30) with $f > 0$. Then the Gaussian actor is given by (35), and the HJB closes in the two coefficient functions f and g , which solve (38)–(39). In particular, the equation for f is linear and independent of g .*

Proof sketch. To see the reduction explicitly, substitute (30) into (29) and collect the coefficient of $(x - \omega)^2$. The terms involving f_p^2/f cancel because the exact wealth–belief cross coefficient satisfies

$$(\sigma b)^\top \Sigma^{-1} (\sigma b) = \|b\|^2. \quad (37)$$

The function f therefore solves the linear parabolic PDE

$$f_t + [a(p) - 2\hat{\rho}(p)^\top b(p)] f_p + \frac{1}{2} \|b(p)\|^2 f_{pp} + \|\hat{\rho}(p)\|^2 f = 0, \quad f(T, p) = 1. \quad (38)$$

The constant coefficient gives

$$g_t + a(p)g_p + \frac{1}{2} \|b(p)\|^2 g_{pp} - \frac{\Lambda}{2} \log \det (\pi \Lambda f(t, p) \Sigma^{-1}) = 0, \quad g(T, p) = 0. \quad (39)$$

Equations (38)–(39) are the exact lower-dimensional coefficient system for the common-covariance benchmark, subject to the regularity and admissibility conditions required by the verification theorem. The equation for f is independent of g , while g is solved after f . $\square$

Table 3 distinguishes structures identified by filtering or control theory from quantities that remain to be estimated numerically. This separation is the operational meaning of *structure-guided* learning in the paper.

Table 3: Control-theoretic structure and learned quantities.

Component	Structure supplied by theory	Quantity estimated numerically
Information state	Posterior belief is sufficient under the filtering model	Regime parameters and recursive belief path
Stage 1 actor	Gaussian family; mean linear in $x - \omega$; covariance proportional to $f \Sigma^{-1}$	Low-dimensional belief/time coefficients and relaxed covariance coefficients
Stage 1 critic	Quadratic dependence on wealth	Coefficient surfaces $f(t, p)$ and $g(t, p)$
Stage 2 policy	QVI continuation/intervention logic and existence of an inaction region	Lower and upper boundary geometry around the Stage 1 center
Transaction costs	Self-financing wealth recursion with proportional cost	Neural boundary parameters under the pathwise objective

The critic uses polynomial coefficient surfaces,

$$\ell_\theta(t, p) = \log f_\theta(t, p) = \sum_{i=0}^m \sum_{j=0}^m \theta_{ij} p^i (T - t)^j, \quad (40)$$

$$g_\theta(t, p) = \sum_{i=0}^{m_g} \sum_{j=0}^{m_g} \tilde{\theta}_{ij} p^i (T - t)^j, \quad (41)$$

and

$$V_\theta(t, x, p; \omega) = \frac{(x - \omega)^2}{\exp(\ell_\theta(t, p))} + g_\theta(t, p) - (\omega - z)^2. \quad (42)$$

The exponential parametrization preserves $f_\theta > 0$. The actor mirrors (35), with a small number of trainable coefficients that allow modest deviations from the exact formula to absorb approximation and filtering errors.

The derivatives used in critic optimization are available analytically. If $\psi_{ij}(t, p) = p^i (T - t)^j$, then

$$\frac{\partial V_\theta}{\partial \theta_{ij}} = -\frac{(x - \omega)^2}{f_\theta(t, p)} \psi_{ij}(t, p), \quad \frac{\partial V_\theta}{\partial \tilde{\theta}_{ij}} = \psi_{ij}(t, p). \quad (43)$$

The displayed polynomial basis does not automatically impose the terminal conditions. If hard terminal consistency is desired, one can instead parameterize

$$\ell_\theta(t, p) = (T - t) \tilde{\ell}_\theta(t, p), \quad g_\theta(t, p) = (T - t) \tilde{g}_\theta(t, p), \quad (44)$$

which guarantees $f_\theta(T, p) = 1$ and $g_\theta(T, p) = 0$; alternatively, the terminal equations can be imposed through supervised residual penalties.

More concretely, the code fixes the wealth dependence through $(x - \omega)$ and learns only low-dimensional belief/time coefficients. A representative actor is

$$\pi_\phi(\mathbf{u} \mid t, x, p) = \mathcal{N}\left(-(x - \omega) \alpha_\phi(p, \partial_p \ell_\theta(t, p)), \frac{\Lambda}{2} f_\theta(t, p) \Sigma_\phi(p)\right), \quad (45)$$

where $\Sigma_\phi(p)$ is constrained to be positive definite. In the exact benchmark, α_ϕ and Σ_ϕ reduce to the coefficients in (35); the relaxed implementation allows small departures without discarding the HJB-implied Gaussian and wealth-linear structure.

A concrete low-dimensional mean map consistent with the exact formula is

$$\alpha_\phi(p, \partial_p \ell_\theta) = (\sigma^\top)^{-1} [\phi_1 p + \phi_2 - \phi_3 p(1 - p) \partial_p \ell_\theta(t, p)], \quad (46)$$

where the vector coefficients ϕ_1, ϕ_2, ϕ_3 learn the filtered myopic and hedging directions. A Cholesky or matrix-exponential parametrization of Σ_ϕ preserves positive definiteness throughout training. The resulting number of actor parameters grows with the asset dimension but not with an unrestricted discretization of the wealth state.

4.3 Policy evaluation and improvement

At each discrete time $t_k = k\Delta t$, the state is $s_k = (t_k, X_k, p_k)$, an action is sampled from the current Gaussian actor, and frictionless wealth is propagated by

$$\mathbf{u}_k \sim \pi_\phi(\cdot \mid s_k), \quad X_{k+1} = X_k + \mathbf{u}_k^\top (\exp(\Delta \log \mathbf{S}_k - r\Delta t) - \mathbf{1}), \quad (47)$$

where the exponential is componentwise. The observed return is then passed to the appropriate common- or regime-dependent-covariance filter.

For a fixed policy, V^π satisfies a linear policy-evaluation PDE. Applying Itô's formula and using that PDE shows that

$$M_t = V^\pi(t, X_t, p_t) - \Lambda \int_0^t \mathcal{H}_s ds \quad (48)$$

is a martingale. Hence

$$V^\pi(t_k, X_k, p_k) = \mathbb{E}_k \left[(X_T - \omega)^2 - (\omega - z)^2 - \Lambda \int_{t_k}^T \mathcal{H}_s ds \right]. \quad (49)$$

This is the continuous-time justification for the terminal Monte Carlo target below; it is not an arbitrary supervised label.

For a simulated episode $\{(t_k, X_k, p_k, \mathbf{u}_k)\}_{k=0}^K$, define the realized cost-to-go

$$G_k = (X_T - \omega)^2 - (\omega - z)^2 - \Lambda \sum_{i=k}^{K-1} \mathcal{H}_i \Delta t. \quad (50)$$

The critic minimizes the Monte Carlo value error

$$\mathcal{L}_{\text{critic}}(\theta) = \frac{1}{2} \mathbb{E}[(V_\theta(t_k, X_k, p_k) - G_k)^2]. \quad (51)$$

The local martingale increment is

$$\Delta M_k = V_\theta(t_{k+1}, X_{k+1}, p_{k+1}) - V_\theta(t_k, X_k, p_k) - \Lambda \mathcal{H}_k \Delta t. \quad (52)$$

In implementation, the actor signal mixes a normalized local increment with the full-horizon tail cost residual

$$r_k^{\text{tail}} = G_k - V_\theta(t_k, X_k, p_k), \quad s_k = (1 - \eta) \text{Norm}(\Delta M_k) + \eta \text{Norm}(r_k^{\text{tail}}). \quad (53)$$

Because s_k is a cost signal, its sign in the policy loss is chosen consistently with minimization. The tail term is an implementation refinement for finite-sample credit assignment; the martingale relation remains the theoretical anchor.

Writing $\text{sg}(\cdot)$ for a stop-gradient operation, a cost-minimizing score-function surrogate is

$$\mathcal{L}_{\text{actor}}(\phi) = \mathbb{E}[\log \pi_\phi(\mathbf{u}_k | t_k, X_k, p_k) \text{sg}(s_k) - \Lambda \mathcal{H}(\pi_\phi(\cdot | t_k, X_k, p_k)) \Delta t]. \quad (54)$$

Positive values of the cost signal reduce the probability of the sampled action under gradient descent. The separate entropy derivative in (54) is required because the entropy depends directly on ϕ and is not recovered by stopping the gradient through s_k . Under a reward convention, both the signal and the optimization sign must be reversed consistently.

The implementation permits one or more critic updates per outer iteration and normalizes the martingale and tail signals within a batch. The reported configurations use one critic step. The pure martingale-difference update is closest to continuous-time theory but was empirically slow and noisy; the mixed signal with weight $\eta = 0.3$ provided a stronger full-horizon correction in the representative runs. This modification is algorithmic rather than a new control theorem.

The multiplier is updated through the sample terminal-wealth constraint,

$$\omega \leftarrow \omega - \alpha_\omega (\bar{X}_T - z). \quad (55)$$

Here $\bar{X}_T$ is the batch estimate of expected terminal wealth, optionally smoothed by an exponential moving average; the reported configurations use smoothing parameter 0.5. Indeed, differentiating the Lagrangian value with respect to ω gives a quantity proportional to $-(\mathbb{E}[X_T] - z)$; the outer step is therefore a dual-ascent update for enforcing the target mean.

Algorithm 1 summarizes the complete Stage 1 loop. Regime parameters are estimated before training, but beliefs are updated recursively inside every episode from observed returns only.

Algorithm 1 Stage 1 HJB-guided actor-critic under partial information

```
1: Estimate  $(\boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}^{(i)}, \lambda_i)$  and initialize  $\theta$ ,  $\phi$ , and  $\omega$ .
2: for outer iteration  $m = 1, \dots, M$  do
3:   Simulate or sample return paths and initialize  $(X_0, p_0)$ .
4:   for  $k = 0, \dots, K - 1$  do
5:     Sample  $\mathbf{u}_k \sim \pi_\phi(\cdot \mid t_k, X_k, p_k)$ .
6:     Propagate  $X_{k+1}$  using (47).
7:     Update  $p_{k+1}$  by the prediction-correction filter using the observed return.
8:   end for
9:   Construct all tail targets  $G_k$  from (50).
10:  Perform one or more critic steps on  $\mathcal{L}_{\text{critic}}$ .
11:  Form  $\Delta M_k$ ,  $r_k^{\text{tail}}$ , and the normalized mixed signal  $s_k$ .
12:  Update the Gaussian actor using (54).
13:  Update  $\omega$  using (55).
14: end for
```

5 Regime Estimation and Filtering in Practice

5.1 Threshold-based labeling

The filter recursion is theory-driven, but its inputs must be estimated. The original heuristic route constructs a bull/bear path from an asset or equal-weight index. If P_t is the index level, the rule switches from bull to bear when

$$P_t \leq P_t^{\text{peak}}(1 - Y_l), \quad (56)$$

and from bear to bull when

$$P_t \geq P_t^{\text{trough}}(1 + Y_u). \quad (57)$$

In the threshold-based configuration, $Y_l = 0.19$ and $Y_u = 0.24$ are fixed hyperparameters rather than cross-validated estimates. This method is transparent but sensitive to threshold selection and can produce unreliable labels in a multi-asset setting.

5.2 Statistical jump model

The current regime-dependent-covariance experiment instead uses an equal-weight return index and constructs a standardized feature vector z_k from rolling means and volatilities, downside deviation, rolling drawdown, short-versus-long mean spreads, and volatility ratios. Every rolling feature (rolling mean, rolling standard deviation, downside deviation, rolling drawdown, and the derived spreads and ratios) is constructed using only return observations available at or before time k ; the subsequent feature standardization (z-scoring), however, is computed from the mean and standard deviation of the full estimation sample and is therefore not itself a causal, online transformation. A two-state statistical jump model jointly estimates regime representatives ϑ_1, ϑ_2 and a persistent label path $s_{1:n}$ by solving a problem of the form

$$\min_{\vartheta_1, \vartheta_2, s_{1:n}} \sum_{k=1}^n \|z_k - \vartheta_{s_k}\|_2^2 + \gamma_J \sum_{k=2}^n \mathbf{1}_{\{s_k \neq s_{k-1}\}}. \quad (58)$$

The first term clusters economically similar feature observations; the jump penalty γ_J suppresses implausibly rapid switching. The reported configuration uses a separate 20-year estimation sample, $\gamma_J = 100$, 10 random initializations, and at most 400 iterations. The estimation sample is separated from the independently generated policy-evaluation paths.

The exact feature windows, feature-normalization statistics, and estimation-sample seed are not documented. The regime-dependent-covariance results are therefore interpreted as exploratory and are not included in the repeated-seed inference.

The discrete SJM is fitted by block coordinate descent, as reviewed by Aydinhan et al. [1]. For a fixed state path, the regime representatives solve

$$\vartheta_j^{(m+1)} = \arg \min_{\vartheta} \sum_{k: s_k^{(m)} = j} \|z_k - \vartheta\|_2^2, \quad (59)$$

so under squared Euclidean loss each representative is the sample mean of the feature vectors assigned to that regime. For fixed representatives, the entire label path is optimized jointly rather than greedily. Define the Bellman cost

$$C_k(j) = \min_{s_{1:k-1}} \left\{ \sum_{r=1}^k \|z_r - \vartheta_{s_r}\|_2^2 + \gamma_J \sum_{r=2}^k \mathbf{1}_{\{s_r \neq s_{r-1}\}} : s_k = j \right\}. \quad (60)$$

It satisfies the dynamic-programming recursion

$$C_1(j) = \|z_1 - \vartheta_j\|_2^2, \quad (61)$$

$$C_k(j) = \|z_k - \vartheta_j\|_2^2 + \min_{i \in \{1, \dots, K\}} \{C_{k-1}(i) + \gamma_J \mathbf{1}_{\{i \neq j\}}\}. \quad (62)$$

Storing the minimizing predecessor at every (k, j) and backtracking from $\arg \min_j C_n(j)$ recovers the globally optimal state sequence conditional on the current representatives. The recursion costs $O(nK^2)$ time and $O(nK)$ memory with full backpointers; for the two-state model used here, its time complexity is linear in the sample length.

Each coordinate-descent step weakly decreases the SJM objective, but the joint problem is nonconvex because labels and representatives are optimized together. The algorithm therefore guarantees only a coordinate-wise stationary solution, not a global optimum. Multiple K-means++-type initializations are used and the solution with the lowest objective is retained. Thus, dynamic programming solves the discrete path subproblem exactly, while random restarts address—but do not eliminate—the global nonconvexity of the complete fitting problem.

Given inferred labels $\hat{I}_k$, the parameters are estimated as follows:

- $\hat{\mu}^{(i)}$ is the annualized mean return within regime i ;
- $\hat{\Sigma}^{(i)}$ is the annualized log-return covariance within regime i ;
- $\hat{\lambda}_1$ and $\hat{\lambda}_2$ are obtained from empirical transition frequencies or, equivalently, inverse average sojourn times.

The jump model is not the online regime predictor used by the controller. It provides a persistent and practically usable segmentation for parameter estimation. Online beliefs are subsequently updated by the Wonham or discrete-time Gaussian Bayes filter, preserving the belief-state HJB architecture.

If N_{ij} is the number of inferred transitions from state i to state j , the empirical transition matrix is

$$\hat{P}_{ij} = \frac{N_{ij}}{\sum_{j'} N_{ij'}}, \quad (63)$$

and the continuous-time switching intensities are obtained from the small-step approximation $\hat{\lambda}_1 \simeq \hat{P}_{12}/\Delta t$ and $\hat{\lambda}_2 \simeq \hat{P}_{21}/\Delta t$, or equivalently from inverse mean sojourn times.

Table 4 reports the SJM-derived parameters used in the regime-dependent run. Regime 1 has positive mean returns and lower covariance, whereas regime 2 has negative mean returns, higher marginal volatility, and higher cross-asset correlation. The estimated switching process is less asymmetric than the true simulation process: it exits regime 1 more frequently and exits regime 2 less rapidly.

Table 4: Statistical jump model results used by the regime-dependent-covariance filter.

Quantity	Regime 1	Regime 2
Annualized mean return	(0.3830, 0.2759)	(−0.3243, −0.1476)
Estimated covariance	$\begin{pmatrix} 0.0485 & 0.0108 \\ 0.0108 & 0.0322 \end{pmatrix}$	$\begin{pmatrix} 0.0574 & 0.0193 \\ 0.0193 & 0.0401 \end{pmatrix}$
Annualized volatilities	(0.220, 0.179)	(0.240, 0.200)
Cross-asset correlation	0.274	0.402
Leaving intensity	$\hat{\lambda}_1 = 0.6801$	$\hat{\lambda}_2 = 1.3310$
Mean sojourn time	1.47 years (370 days)	0.75 years (189 days)

The stationary mass implied by the estimated rates is

$$\hat{p}_1 = \frac{\hat{\lambda}_2}{\hat{\lambda}_1 + \hat{\lambda}_2} \approx 0.662. \quad (64)$$

This estimate is materially below the true simulation value 0.889, which helps explain the noisier belief and learning dynamics in the regime-dependent experiment.

At discrete times, prediction under (2) is

$$p_{k|k-1} = p_{k-1}(1 - \lambda_1 \Delta t) + (1 - p_{k-1})\lambda_2 \Delta t. \quad (65)$$

The predicted weights are combined with regime-wise Gaussian likelihoods and normalized in log space. This distinction between *parameter estimation* and *online filtering* permits richer offline regime identification without replacing the posterior state required by the control model.

6 Regime-Dependent Covariance Extension

6.1 Discrete-time Gaussian filtering layer

In the extension, both drift and covariance depend on the regime:

$$\Delta \log \mathbf{S}_k \mid I_k = i \sim \mathcal{N} \left(\left(\boldsymbol{\mu}^{(i)} - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma}^{(i)}) \right) \Delta t, \boldsymbol{\Sigma}^{(i)} \Delta t \right). \quad (66)$$

Let $\phi_i(y_k)$ denote the Gaussian density in (66). After the Markov prediction (13), Bayes' rule gives

$$p_k = \frac{p_{k|k-1} \phi_1(y_k)}{p_{k|k-1} \phi_1(y_k) + (1 - p_{k|k-1}) \phi_2(y_k)}. \quad (67)$$

In computation, the numerator and denominator are evaluated through normalized log weights. The Bayes correction in (67) is exact conditional on the specified Gaussian emission model and the predicted probability. The prediction (13), however, is the first-order Euler discretization of the continuous-time Markov chain; an exactly discretized alternative would use the transition matrix $\exp(Q\Delta t)$.

The continuous-time separation structure is more delicate because the hidden state affects the observation quadratic variation as well as its drift. In particular, the common- σ innovation construction (8) cannot be reused without a new filtering theorem. The extension therefore separates a discrete-time prediction–correction filtering layer from an approximate control layer.

6.2 Moment-matched effective diffusion

Write $m_i = \boldsymbol{\mu}^{(i)} - \frac{1}{2} \text{diag}(\boldsymbol{\Sigma}^{(i)})$. Conditional on a prior belief p , the one-step mixture has

$$\mathbb{E}[\Delta \log \mathbf{S}_k \mid p] = (pm_1 + (1 - p)m_2) \Delta t, \quad (68)$$

and

$$\begin{aligned} \text{Cov}(\Delta \log \mathbf{S}_k \mid p) &= \left[p\boldsymbol{\Sigma}^{(1)} + (1 - p)\boldsymbol{\Sigma}^{(2)} \right] \Delta t \\ &\quad + p(1 - p)(m_1 - m_2)(m_1 - m_2)^\top (\Delta t)^2. \end{aligned} \quad (69)$$

Thus the belief-averaged covariance is the leading-order $O(\Delta t)$ conditional covariance; the between-regime drift-mixture term is $O((\Delta t)^2)$. For control, the implementation defines

$$\hat{\mathbf{b}}(p) = p(\boldsymbol{\mu}^{(1)} - r\mathbf{1}) + (1 - p)(\boldsymbol{\mu}^{(2)} - r\mathbf{1}), \quad \bar{\boldsymbol{\Sigma}}(p) = p\boldsymbol{\Sigma}^{(1)} + (1 - p)\boldsymbol{\Sigma}^{(2)}. \quad (70)$$

The controlled wealth process is approximated by the moment-matched diffusion

$$dX_t \approx \mathbf{u}_t^\top \hat{\mathbf{b}}(p_t) dt + \mathbf{u}_t^\top \bar{\boldsymbol{\sigma}}(p_t) d\mathbf{B}_t, \quad \bar{\boldsymbol{\sigma}}(p) \bar{\boldsymbol{\sigma}}(p)^\top = \bar{\boldsymbol{\Sigma}}(p). \quad (71)$$

Equation (71) is a single effective diffusion that matches the leading conditional drift and covariance. It is not the exact separated wealth dynamics of the original partially observed continuous-time model.

6.3 Approximate HJB and Gaussian actor

To make the approximation explicit, posit effective belief dynamics

$$dp_t = \alpha(p_t) dt + \beta(p_t)^\top dW_t, \quad q(p) = \|\beta(p)\|^2, \quad (72)$$

and let $\tilde{c}(p) \in \mathbb{R}^d$ denote the effective covariance between wealth noise per unit action and belief noise. The formal approximate HJB is

$$0 = v_t + \alpha(p)v_p + \frac{1}{2}q(p)v_{pp} + \min_{\pi} \int_{\mathbb{R}^d} \left[\frac{1}{2} \mathbf{u}^\top \bar{\Sigma}(p) \mathbf{u} v_{xx} + \mathbf{u}^\top (\hat{\mathbf{b}}(p)v_x + \tilde{c}(p)v_{xp}) + \Lambda \ln \pi(\mathbf{u}) \right] \pi(\mathbf{u}) d\mathbf{u}. \quad (73)$$

The Hamiltonian is still quadratic in $\mathbf{u}$. Therefore, if $v_{xx} > 0$,

$$\pi^*(\cdot | t, x, p) = \mathcal{N} \left(-\bar{\Sigma}(p)^{-1} \frac{\hat{\mathbf{b}}(p)v_x + \tilde{c}(p)v_{xp}}{v_{xx}}, \frac{\Lambda}{v_{xx}} \bar{\Sigma}(p)^{-1} \right). \quad (74)$$

With the same wealth-quadratic ansatz (30), let $\ell = f_p/f$. The policy becomes

$$\pi^* = \mathcal{N} \left(-(x - \omega) \bar{\Sigma}(p)^{-1} [\hat{\mathbf{b}}(p) - \tilde{c}(p)\ell(t, p)], \frac{\Lambda f(t, p)}{2} \bar{\Sigma}(p)^{-1} \right). \quad (75)$$

Collecting the $(x - \omega)^2$ terms in (73) yields the approximate coefficient equation

$$0 = f_t + \left[\alpha(p) - 2\hat{\mathbf{b}}(p)^\top \bar{\Sigma}(p)^{-1} \tilde{c}(p) \right] f_p + \frac{1}{2}q(p)f_{pp} + f\hat{\mathbf{b}}(p)^\top \bar{\Sigma}(p)^{-1} \hat{\mathbf{b}}(p) + [\tilde{c}(p)^\top \bar{\Sigma}(p)^{-1} \tilde{c}(p) - q(p)] \frac{f_p^2}{f}, \quad f(T, p) = 1. \quad (76)$$

The constant term gives

$$g_t + \alpha(p)g_p + \frac{1}{2}q(p)g_{pp} - \frac{\Lambda}{2} \log \det (\pi \Lambda f(t, p) \bar{\Sigma}(p)^{-1}) = 0, \quad g(T, p) = 0. \quad (77)$$

Equation (76) reveals the precise difference from the theorem-level benchmark. In the common-covariance case,

$$\tilde{c} = \sigma b, \quad \bar{\Sigma} = \sigma \sigma^\top, \quad q = \|b\|^2, \quad (78)$$

so $\tilde{c}^\top \bar{\Sigma}^{-1} \tilde{c} = q$ and the nonlinear f_p^2/f term cancels, recovering (38). Under the regime-dependent approximation, this identity is not established; the additional nonlinear term must be retained unless the effective coefficients are deliberately constrained to satisfy it.

In the reported implementation, α , β , and the wealth–belief cross coefficient $\tilde{c}$ are not separately identified from a continuous-time regime-dependent-covariance filtering model. Beliefs are updated by the Euler-prediction and exact Gaussian Bayes-correction recursion in (67), and the belief-averaged covariance $\bar{\Sigma}(p)$ is imposed explicitly in the actor. The remaining effects represented formally by α , β , and $\tilde{c}$ are absorbed into the relaxed low-dimensional actor and critic coefficients. Consequently, (73)–(76) motivate the Gaussian architecture and coefficient dependence; they are not PDEs solved directly in the reported experiment.

6.4 Exact and approximate components

- **Exact conditional correction:** the Gaussian Bayes update in (67), conditional on the predicted probability and specified emission densities.
- **Time-discretization convention:** the reported Markov prediction uses the first-order formula (13), rather than the exact matrix exponential $\exp(Q\Delta t)$.
- **Moment-matched approximation:** replacement of the posterior Gaussian mixture by (71).
- **Structurally preserved:** quadratic Hamiltonian, Gaussian exploratory policy, wealth-quadratic ansatz, and low-dimensional coefficient representation.
- **Implementation convention:** $\bar{\Sigma}(p)$ and the discrete Bayesian belief recursion are explicit, while the continuous-time coefficients $(\alpha, \beta, \tilde{c})$ are not estimated separately.
- **Not yet proved:** an exact continuous-time filter with regime-dependent covariance, a separation principle, identification of $\tilde{c}$, admissibility and well-posedness of (76)–(77), and a verification theorem.

Remark 1. *The regime-dependent results should therefore be interpreted as a discretized prediction–correction filter combined with a theory-guided approximate controller. The numerical experiment tests whether this architecture remains learnable; it does not establish optimality for the original continuous-time partially observed model.*

7 Stage 2: Transaction-Cost-Aware No-Trade Learning

7.1 Transaction-cost QVI benchmark

With transaction costs, the current position affects all future trading costs. The observable state must therefore be enlarged from (t, x, p) to $(t, x, p, \mathbf{w})$. In the no-trade region, consider the weights-based diffusion

$$dX_t = X_t \bar{\mu}(p_t, \mathbf{w}_t) dt + X_t \bar{\sigma}(p_t, \mathbf{w}_t)^\top d\hat{\mathbf{B}}_t, \quad (79)$$

$$dp_t = \mu_p(p_t) dt + \beta(p_t)^\top d\hat{\mathbf{B}}_t, \quad (80)$$

$$d\mathbf{w}_t = b_w(p_t, \mathbf{w}_t) dt + D(p_t, \mathbf{w}_t) d\hat{\mathbf{B}}_t. \quad (81)$$

The continuation generator $\mathcal{L}$ contains all state covariations generated by the common innovation. In particular,

$$a_{Xp} = x \bar{\sigma}^\top \beta, \quad a_{Xw} = x D \bar{\sigma}, \quad a_{pw} = D \beta, \quad a_{ww} = D D^\top. \quad (82)$$

Let $\mathcal{M}$ be the intervention operator. For a cost-minimization convention, the transaction-cost dynamic program is the QVI

$$\min\{-V_t - \mathcal{L}V, \mathcal{M}V - V\} = 0. \quad (83)$$

The continuation region satisfies $-V_t - \mathcal{L}V = 0$ and $V < \mathcal{M}V$; intervention occurs on the contact set $V = \mathcal{M}V$. For an approximate critic, a negative obstacle residual $\mathcal{M}_\phi V_\theta - V_\theta$ indicates that the candidate intervention has lower cost than continuation.

Motivated by the Stage 1 terminal condition, consider

$$V(t, x, p, \mathbf{w}) = A(t, p, \mathbf{w})(x - \omega)^2 + B(t, p, \mathbf{w})(x - \omega) + C(t, p, \mathbf{w}), \quad (84)$$

with

$$A(T, p, \mathbf{w}) = 1, \quad B(T, p, \mathbf{w}) = 0, \quad C(T, p, \mathbf{w}) = -(\omega - z)^2. \quad (85)$$

Writing $y = x - \omega$, every derivative of (84) is polynomial in y . Consequently,

$$-V_t - \mathcal{L}V = \Phi_2(t, p, \mathbf{w})y^2 + \Phi_1(t, p, \mathbf{w})y + \Phi_0(t, p, \mathbf{w}), \quad (86)$$

and continuation requires $\Phi_2 = \Phi_1 = \Phi_0 = 0$. Unlike frictionless Stage 1, the linear coefficient B is generally generated by the wealth-proportional dynamics and cannot be fixed at zero away from maturity. Explicit forms of the three coefficients are collected in Appendix A.

7.2 Intervention operator and quadratic closure

Let a general intervention actor choose

$$\mathbf{w}' = \Gamma_\phi(t, p, \mathbf{w}), \quad (87)$$

and let $\kappa(\mathbf{w}, \mathbf{w}')$ be the proportional fraction of wealth lost in trading. Then

$$x^+ = x(1 - \kappa), \quad (\mathcal{M}_\phi V)(t, x, p, \mathbf{w}) = V(t, x^+, p, \Gamma_\phi(t, p, \mathbf{w})). \quad (88)$$

Since $x^+ - \omega = (1 - \kappa)y - \kappa\omega$, the intervention operator preserves the quadratic family whenever Γ_ϕ and κ are independent of x conditional on weights:

$$\mathcal{M}_\phi V = M_2^\phi y^2 + M_1^\phi y + M_0^\phi, \quad (89)$$

where, with $A^\Gamma = A(t, p, \Gamma_\phi)$ and similarly for B^Γ, C^Γ ,

$$M_2^\phi = A^\Gamma(1 - \kappa)^2, \quad (90)$$

$$M_1^\phi = -2A^\Gamma \kappa \omega(1 - \kappa) + B^\Gamma(1 - \kappa), \quad (91)$$

$$M_0^\phi = A^\Gamma \kappa^2 \omega^2 - B^\Gamma \kappa \omega + C^\Gamma. \quad (92)$$

This closure suggests a QVI-guided critic for (A, B, C) and an intervention actor Γ_ϕ .

To retain Stage 1-style low dimensionality in the exploratory QVI design, introduce an exogenous normalized reference map $\mathbf{w}^{\text{ref}}(t, p)$, for example the Stage 1 policy evaluated at a reference wealth level, and define

$$\tau = T - t, \quad \delta = \mathbf{w} - \mathbf{w}^{\text{ref}}(t, p). \quad (93)$$

A structured critic can use even powers of $\|\delta\|_{M(p)}$ for A and C , and odd terms for B ; a corresponding intervention actor can contract δ toward zero. Appendix B gives the proposed basis and residual losses.

This reference-map convention is an architectural approximation. The implemented Stage 1 center weight is generally wealth dependent because $\mathbf{w}^*(t, x, p) = \mathbf{u}^*(t, x, p)/x$. No exact QVI closure is claimed after replacing that map by the wealth-independent reference $\mathbf{w}^{\text{ref}}(t, p)$.

7.3 Exploratory QVI-guided Stage 2 RL design

This subsection records the exploratory actor–critic design considered before adopting Direct Boundary policy optimization. It is a work-in-progress numerical formulation, not a completed verification result. The purpose is to preserve in Stage 2 the same principle used successfully in Stage 1: derive a restricted critic and actor from the control equation before introducing learning.

The critic learns the three coefficient surfaces in (84),

$$A_\theta(\tau, p, \delta), \quad B_\theta(\tau, p, \delta), \quad C_\theta(\tau, p, \delta), \quad (94)$$

where $\tau = T - t$ and $\delta = \mathbf{w} - \mathbf{w}^{\text{ref}}(t, p)$. Positivity of A_θ can be imposed exponentially, while low-order even and odd center-relative bases reflect the quadratic continuation equations. The terminal condition may be enforced either exactly through the parametrization or softly through a terminal residual penalty.

The intervention actor proposes a post-trade target through

$$\Gamma_\phi(t, p, \mathbf{w}) = \mathbf{w} + G_\phi(\tau, p, \delta) = \mathbf{w}^{\text{ref}}(t, p) + \delta + G_\phi(\tau, p, \delta). \quad (95)$$

A low-dimensional radial–directional decomposition is

$$G_\phi(\tau, p, \delta) = -s_\phi(\tau, p, \delta)\delta + R_\phi(\tau, p, \delta), \quad 0 \leq s_\phi \leq 1. \quad (96)$$

With $R_\phi \equiv 0$, $s_\phi = 0$ leaves the position unchanged and $s_\phi = 1$ moves it to the normalized reference center. A simple structured shrinkage factor is

$$s_\phi = \text{sigmoid} \left(\eta_0(\tau, p) + \eta_2(\tau, p) \|\delta\|_{M(p)}^2 + \eta_4(\tau, p) \|\delta\|_{M(p)}^4 \right). \quad (97)$$

The directional correction R_ϕ can be introduced only if the radial model is too restrictive. A final differentiable projection or normalization layer enforces $\Gamma_\phi(t, p, \mathbf{w}) \in \mathcal{W}$ under box, simplex, or leverage constraints.

For the cost-minimization convention, define

$$R_{\text{cont}} = -V_{\theta,t} - \mathcal{L}V_\theta, \quad R_{\text{obs}} = \mathcal{M}_\phi V_\theta - V_\theta. \quad (98)$$

The exact QVI requires both residuals to be nonnegative and complementary. A negative value of R_{obs} means that the candidate post-trade value is lower than the current continuation value and therefore signals an intervention opportunity. A differentiable soft trigger is

$$\alpha_{\theta,\phi}(t, p, \mathbf{w}) = \text{sigmoid}(-\zeta_{\text{tr}} R_{\text{obs}}(t, p, \mathbf{w})), \quad (99)$$

although two alternatives are possible: a separate trade/no-trade score head, or always evaluating (95) and interpreting $\Gamma_\phi \approx \mathbf{w}$ as no trade.

A schematic critic objective is

$$\begin{aligned} \mathcal{L}_{\text{critic}}^{\text{QVI}} = & \mathcal{L}_{\text{terminal}} + \lambda_c \mathbb{E}[(R_{\text{cont}}^-)^2] + \lambda_o \mathbb{E}[(R_{\text{obs}}^-)^2] \\ & + \lambda_{co} \mathbb{E}[(R_{\text{cont}} R_{\text{obs}})^2], \end{aligned} \quad (100)$$

where $r^- = \min(r, 0)$. The actor can combine the simulated terminal objective with feasibility and trigger regularization,

$$\mathcal{L}_{\text{actor}}^{\text{QVI}} = \mathcal{L}_{\text{pathwise}} + \lambda_\Gamma \mathcal{L}_{\text{feas}} + \lambda_{\text{tr}} \mathcal{L}_{\text{trigger}}. \quad (101)$$

In principle, this design allows pure QVI-residual fitting, pure pathwise policy optimization, or a hybrid. A practical future route is to initialize the intervention actor from a trained Direct Boundary policy and then introduce the QVI residuals gradually as a curriculum, rather than learning the critic, trigger, and intervention map simultaneously from scratch.

7.4 Direct Boundary policy optimization for Stage 2

The QVI is the natural theoretical benchmark for the transaction-cost problem. Under the optimization scheme considered here, joint learning of the QVI residual and intervention map did not converge reliably. Stage 2 therefore evaluates Direct Boundary policy optimization, which directly optimizes a restricted no-trade policy through simulated wealth paths. The main sources of numerical difficulty were:

- the state dimension increases through the current weight vector $\mathbf{w}$;

- the minimum in (83) creates a non-smooth continuation/intervention boundary;
- intervention samples are sparse and the trigger can collapse to always-trade or never-trade behavior;
- critic errors enter both the continuation residual and the intervention target $\mathcal{M}_\phi V$;
- the terminal mean–variance signal is delayed, while filtering and transaction costs add further pathwise noise.

Direct Boundary policy optimization uses pathwise gradient descent on the terminal objective, without a separate learned transaction-cost critic or value function.

The distinction between the theoretical benchmark and the reported objective is important. The QVI above retains the Stage 1 mean–variance terminal condition, whereas the implemented Direct Boundary policy is trained using the terminal utility and regularization objective in (113). Stage 2 is therefore not claimed to solve the exact transaction-cost extension of the Stage 1 problem. The frozen Stage 1 policy is used as a state-dependent structural feature and advanced start, and all reported Stage 2 policy comparisons are interpreted under the separate transaction-cost-aware objective.

7.5 From a frictionless center to an inaction region

Let the implemented Stage 1 center weight be

$$\mathbf{w}_k^* = \mathbf{w}^*(t_k, x_k, p_k) := \frac{\mathbf{u}^*(t_k, x_k, p_k)}{x_k}, \quad x_k > 0. \quad (102)$$

Because the risky-dollar actor is proportional to $x_k - \omega$, this normalized center is generally wealth dependent. Under proportional costs, mechanically rebalancing to $\mathbf{w}_k^*$ at every step can destroy value through turnover. Motivated by the control-limit result of Taksar et al. [13], Stage 2 learns a no-trade region around this wealth-dependent center.

The Stage 2 pre-trade state is

$$o_k = [\tau_k, x_k^-, p_k, \mathbf{w}_k^-, \mathbf{w}_k^*, \mathbf{w}_k^- - \mathbf{w}_k^*], \quad \tau_k = \max(0, 1 - t_k/T). \quad (103)$$

A feedforward network predicts positive lower and upper gaps,

$$\mathbf{g}_k^\ell = \text{softplus}(h_\phi^\ell(o_k)) + \varepsilon, \quad \mathbf{g}_k^u = \text{softplus}(h_\phi^u(o_k)) + \varepsilon, \quad (104)$$

and defines

$$\underline{\mathbf{w}}_k = \mathbf{w}_k^* - \mathbf{g}_k^\ell, \quad \overline{\mathbf{w}}_k = \mathbf{w}_k^* + \mathbf{g}_k^u. \quad (105)$$

Apply the componentwise no-trade-band projection

$$\mathbf{w}_k^{\text{band}} = \min(\max(\mathbf{w}_k^-, \underline{\mathbf{w}}_k), \overline{\mathbf{w}}_k). \quad (106)$$

For the training recursion, the post-trade target is $\mathbf{w}_k^+ = \mathbf{w}_k^{\text{band}}$ directly: the network is trained by differentiating through (106) and the wealth recursion below, and high gross leverage is discouraged only by the soft L_1 regularizer $\mathcal{L}_{gl}$ in (113) (coefficient $\lambda_{gl} = 10^{-5}$), not by a hard constraint. A hard gross-leverage cap,

$$\begin{aligned} \mathbf{w}_k^{+, \text{eval}} &= \Pi_{\mathcal{W}}(\mathbf{w}_k^{\text{band}}), \quad \mathcal{W} = \{\mathbf{w} : \|\mathbf{w}\|_1 \leq L_{\max}\}, \quad L_{\max} = 1.5, \\ \Pi_{\mathcal{W}}(\mathbf{w}) &= \begin{cases} \mathbf{w} & \|\mathbf{w}\|_1 \leq L_{\max} \\ \mathbf{w} \cdot L_{\max} / \|\mathbf{w}\|_1 & \|\mathbf{w}\|_1 > L_{\max} \end{cases}, \end{aligned} \quad (107)$$

is instead applied only as a post-hoc *evaluation-time* constraint, identically (same $L_{\max} = 1.5$, same proportional-rescaling rule) to every reported Stage 2 comparator – Center+DNNBand and all three center/equal-weight baselines – so that comparisons are not confounded by an asymmetric constraint. It is not part of the training objective or the training-time wealth recursion. Thus no trade occurs inside the band whenever the current pre-trade position is also feasible; outside the band, the portfolio is moved to the nearest boundary. At evaluation time only, a position exceeding the leverage cap is additionally rescaled proportionally toward the origin until its L_1 norm equals $L_{\max}$. Unlike a symmetric fixed-width rule, the network permits asymmetric, time-varying, belief-dependent boundaries. The risky-weight vector excludes cash, so the discounted cash weight is $1 - \mathbf{1}^\top \mathbf{w}_k$.

7.6 Wealth recursion and direct objective

To make the trading-time convention explicit, let $(x_k^-, \mathbf{w}_k^-)$ denote pre-trade discounted wealth and risky weights at time t_k . The band rule in (106) produces the post-trade target $\mathbf{w}_k^+ = \mathbf{w}_k^{\text{band}}$ used in training; at evaluation time, $\mathbf{w}_k^+$ is additionally replaced by $\mathbf{w}_k^{+, \text{eval}}$ from (107) for every reported method. With proportional transaction-cost rate κ , define

$$\text{TC}_k = \kappa x_k^- \|\mathbf{w}_k^+ - \mathbf{w}_k^-\|_1, \quad (108)$$

$$x_k^+ = x_k^- - \text{TC}_k, \quad (109)$$

$$\mathbf{u}_k^+ = x_k^+ \mathbf{w}_k^+, \quad (110)$$

$$x_{k+1}^- = x_k^+ + (\mathbf{u}_k^+)^{\top} R_{k+1}^{\text{disc}}, \quad (111)$$

$$\mathbf{u}_{k+1}^- = \mathbf{u}_k^+ \odot (\mathbf{1} + R_{k+1}^{\text{disc}}), \quad (112)$$

where R_{k+1}^{disc} is the vector of discounted simple risky returns over $(t_k, t_{k+1}]$ and $\odot$ denotes componentwise multiplication. The next pre-trade weight is $\mathbf{w}_{k+1}^- = \mathbf{u}_{k+1}^- / x_{k+1}^-$. Elsewhere in the Direct Boundary description, $(x_k, \mathbf{w}_k)$ denotes this pre-trade state $(x_k^-, \mathbf{w}_k^-)$. This notation ensures that transaction costs are charged on the actual change from current pre-trade holdings to the selected post-trade target, rather than on the previous period's nominal post-trade dollars.

The training and evaluation code uses this same pre-/post-trade timing convention throughout, with the evaluation seed recorded for every training seed (Table 6). The network is trained by differentiating through (106) and (112) (with $\mathbf{w}_k^+ = \mathbf{w}_k^{\text{band}}$, i.e. without the evaluation-time leverage cap (107)). The implemented objective, minimized over an episode of N steps and averaged over the 128 simulated episodes per training iteration, is

$$\mathcal{L}_{\text{band}}(\phi) = -\lambda_U (\mathbb{E}[U(x_T)] - s_U) + \lambda_{to} \mathcal{L}_{to} + \lambda_g \mathcal{L}_g + \lambda_{gl} \mathcal{L}_{gl}, \quad (113)$$

where

$$\mathcal{L}_{to} = \mathbb{E} \left[\frac{1}{N} \sum_{k=0}^{N-1} \|\mathbf{w}_k^+ - \mathbf{w}_k^-\|_1 \right], \quad (114)$$

$$\mathcal{L}_g = \mathbb{E} \left[\frac{1}{N} \sum_{k=0}^{N-1} \frac{1}{2} (\|\mathbf{g}_k^{\ell}\|_2^2 + \|\mathbf{g}_k^u\|_2^2) \right], \quad (115)$$

$$\mathcal{L}_{gl} = \mathbb{E} \left[\frac{1}{N} \sum_{k=0}^{N-1} \|\mathbf{w}_k^+\|_1 \right], \quad (116)$$

the outer $\mathbb{E}[\cdot]$ denotes the average over simulated episodes, $N = 252$ is the number of daily steps in the one-year horizon, and the factor $\frac{1}{2}$ in $\mathcal{L}_g$ is the average (not sum) of the squared lower- and upper-gap norms over the two risky assets. In the implemented code, λ_U is the scalar `utility_scale` and s_U is an additive shift `utility_shift`; for the reported runs $\lambda_U = 1$ and $s_U = 0$, so the term reduces to $-\mathbb{E}[U(x_T)]$ as in a standard terminal-utility objective. The physical proportional transaction cost is already deducted from wealth in (112). The additional turnover term $\mathcal{L}_{to}$ is therefore a deliberate governance and training regularizer—and a simple proxy for unmodeled market impact—rather than a second accounting deduction of the same transaction cost. The gap term $\mathcal{L}_g$ discourages uninformatively wide bands; the gross-leverage term $\mathcal{L}_{gl}$ is the only leverage control active during training; the separate, evaluation-only hard cap is discussed above and in Section 8.1.

In the reported configuration, U is logarithmic utility, $U(x) = \log(\max(x, \epsilon))$ with positivity floor $\epsilon = 10^{-10}$ applied directly inside the utility evaluation (not as a separate bankruptcy-penalty term); no additional nonpositive-wealth penalty is added to the loss. Although the implementation exposes the parameters `utility_gamma` and `gamma_risk`, neither is active under the reported logarithmic-utility and direct-gap specification. Neither parameter therefore affects the reported policy or objective. The loss coefficients used are $\lambda_{to} = 5 \times 10^{-4}$ (`turnover_coef`), $\lambda_g = 10^{-5}$ (`gap_l2_coef`), and $\lambda_{gl} = 10^{-5}$ (`gross_lev_coef`), as recorded in Table 6. The checkpoint used for evaluation is selected, every 25 training iterations, as the iteration with the highest mean validation utility $\mathbb{E}[U(x_T)]$ over 64 held-out validation paths generated independently of both the training and the 1000-path evaluation set. This validation utility is computed by the same band-only simulator used in training, i.e. without the evaluation-time leverage cap (107): the checkpoint-selection criterion and the final reported evaluation metrics are therefore not evaluated under an identical policy, since only the latter additionally applies the hard leverage cap on top of the learned band.

Typical terminal utilities include logarithmic, square-root, and CRRA utility, all of which require an explicit treatment of nonpositive wealth.

Direct Boundary policy optimization is related to the HJB advanced-start strategy of Mulvey et al. [9]: a tractable frictionless solution supplies information that makes the transaction-cost problem easier to learn. The QVI derivation above explains why a no-trade region and center-relative coordinates are natural, but the reported

optimizer directly trains the restricted boundary policy class because the attempted QVI critic/intervention actor was unstable. The QVI remains the theoretical benchmark, but the evaluated boundary policy is not its numerical solution.

7.7 Unified training procedure

Algorithm 2 Two-stage structure-guided learning procedure

- 1: Label regime episodes from simulated or historical return paths: drawdown-threshold heuristic labeling for the common-covariance track reported in Section 8.3, or the statistical jump model of Section 5.2 for the regime-dependent-covariance extension.
 - 2: Estimate $\boldsymbol{\mu}^{(i)}$, $\boldsymbol{\Sigma}^{(i)}$ (common or regime-dependent), and λ_i from the inferred labels.
 - 3: Initialize the belief filter, Stage 1 critic, Gaussian actor, and multiplier ω .
 - 4: **for** each Stage 1 training iteration **do**
 - 5: Simulate market paths and update beliefs from observed returns.
 - 6: Sample actions from the structured Gaussian policy and propagate frictionless wealth.
 - 7: Update the critic, actor, and multiplier using the cost-to-go and martingale signals.
 - 8: **end for**
 - 9: Freeze the Stage 1 center map $\boldsymbol{w}^*(t, x, p)$.
 - 10: **for** each Stage 2 training iteration **do**
 - 11: Simulate common market and belief paths and compute the center path.
 - 12: Predict no-trade gaps via Direct Boundary policy optimization and apply the projection rule.
 - 13: Propagate transaction-cost-aware wealth and minimize (113).
 - 14: **end for**
 - 15: Evaluate Stage 1 and Stage 2 policies on common independently generated simulation paths.
-

8 Numerical Study

8.1 Experimental design

The reported experiments use a two-regime, two-risky-asset environment with one trading year, daily time steps, initial discounted wealth $x_0 = 1$, initial belief $p_0 = 0.5$, risk-free rate $r = 0.01$, and Stage 1 target $z = 1.2$. The common-covariance and regime-dependent covariance Stage 1 models are trained for 9000 iterations. The latter uses the statistical jump model with jump penalty 100, 10 random initializations, and at most 400 iterations.

The transaction-cost experiment uses the common-covariance Stage 1 center, estimated-parameter filtering, proportional transaction cost $\kappa = 0.002$, and 1000 common evaluation paths. The leverage cap of 1.5 is applied to the target portfolio weight at each evaluation-time rebalancing decision for every reported Stage 2 comparator; it is an evaluation-side constraint and is unrelated to the Stage 1 actor’s own component-wise action bound (Table 5), which is not enforced during Stage 2 training or evaluation. Under (112), $\kappa = 0.002$ represents 20 basis points per unit of absolute pre-to-post-trade risky-weight turnover, scaled by pre-trade wealth. Reported comparators include daily and monthly center rebalancing, monthly-rebalanced equal weight, and Center+DNNBand.

Evaluation metric definitions. The following Stage 2 evaluation metrics are computed exactly as implemented in `poemv_rs/stage2_eval.py`, over $N = 252$ daily steps per one-year path:

- **Net terminal wealth** is the realized, transaction-cost-net terminal discounted wealth x_N produced by the recursion (112) along the actually simulated (cost-aware) trade path.
- **Gross terminal wealth** is a parallel bookkeeping ledger that accumulates the identical per-step realized investment profit-and-loss $(\boldsymbol{u}_k^+)^{\top} R_{k+1}^{\text{disc}}$ along the *same* realized post-trade dollar positions $\boldsymbol{u}_k^+$ as net wealth, but never debits the transaction cost TC_k . It is therefore not a counterfactual re-simulation under a hypothetical zero-cost policy (trade sizes are not rescaled for a larger fee-free capital base), and it does not credit the foregone investment returns that the transaction-cost dollars would have earned had they remained invested; it is an accounting decomposition of realized wealth only. This is discussed further alongside Table 11.
- **Cumulative transaction cost** is defined as (gross terminal wealth – net terminal wealth), i.e. by construction from the two quantities above rather than as an independently accumulated series. This identity is an accounting definition rather than a counterfactual no-cost re-simulation.

- **Turnover** at step k is $\|\mathbf{w}_k^+ - \mathbf{w}_k^-\|_1$ (weight-space, post-leverage-cap at evaluation time). The reported turnover statistic (e.g. 0.0397 for Center+DNNBand) is the *per-step time average* over the 252 steps of a path, further averaged over paths and training seeds – it is neither a cumulative sum over the horizon nor an average conditioned on trading days only.
- **Number of trades** counts steps with turnover exceeding a fixed tolerance, $\sum_{k=0}^{N-1} \mathbf{1}\{\|\mathbf{w}_k^+ - \mathbf{w}_k^-\|_1 > 10^{-9}\}$.
- **Mean band width** at step k is the two-asset *average* (not sum) of $\bar{\mathbf{w}}_k - \underline{\mathbf{w}}_k$; the reported statistic (e.g. 1.740 for Center+DNNBand) is this per-step asset-averaged quantity further averaged over the 252 steps of a path, then over paths, then over training seeds.
- **Target achievement probability** $\mathbb{P}(X_T \geq z)$ is the empirical fraction of evaluation paths whose terminal wealth (net terminal wealth for Stage 2, terminal wealth for Stage 1) meets or exceeds the target $z = 1.2$.

Baseline comparator definitions. All comparators below are subject to the same evaluation-time leverage cap (107) as Center+DNNBand, applied identically after each comparator’s own target-weight rule.

- **Equal weight (Stage 1 and Stage 2 monthly variant)** holds the fixed weight vector $\mathbf{w}_{\text{ew}} = (1/d, \dots, 1/d)$ (here $d = 2$, so $(0.5, 0.5)$) in every risky asset, independent of the belief or wealth state. It is rebalanced back to $\mathbf{w}_{\text{ew}}$ at the comparator’s rebalancing frequency (daily for the Stage 1 evaluation reported in Table 9; monthly, i.e. every 21 trading days, for the Stage 2 “Equal weight, monthly” comparator in Table 11).
- **Minimum variance (Stage 1 only)** holds the fixed global-minimum-variance weight vector $\mathbf{w}_{\text{gmV}} \propto \bar{\Sigma}^{-1} \mathbf{1}$ (renormalized so $\mathbf{1}^\top \mathbf{w}_{\text{gmV}} = 1$), computed once from the pseudo-inverse of the unconditional (stationary-mixture) covariance matrix implied by the true simulation parameters. Like equal weight, it does not depend on the belief state and is rebalanced back to this fixed target at the comparator’s rebalancing frequency.
- **CenterOnly, daily / monthly (Stage 2)** rebalance mechanically, with no no-trade band, to the frozen Stage 1 center weight $\mathbf{w}_k^* = \mathbf{u}^*(t_k, x_k, p_k)/x_k$ of (102) evaluated at the current state, at the stated frequency (every step for “daily”; every 21 steps for “monthly”); between rebalancing dates the position is left untraded. This isolates the value of the Stage 1 center itself, evaluated under the same transaction-cost and leverage-cap conventions as Center+DNNBand, without the learned no-trade boundary.

For Tables 9 and 10, mean shortfall is defined by

$$\text{MSF} := \mathbb{E}[(z - X_T)^+]. \quad (117)$$

Table 5: Selected Stage 1 configuration parameters for the common-covariance and regime-dependent-covariance tracks. Optimizer and actor-mixing hyperparameters (rows below the double rule) are shared between the two tracks and are not independently re-tuned per track.

Item	Common covariance	Regime-dependent covariance
Horizon / step	1 year / 1/252	1 year / 1/252
Initial state	$(x_0, p_0) = (1, 0.5)$	$(x_0, p_0) = (1, 0.5)$
Target / entropy	$z = 1.2, \Lambda = 1$	$z = 1.2, \Lambda = 1$
Iterations	9000	9000
Episodes per outer iteration	16	not fixed to a single value for this exploratory track
Covariance control	common Σ	$\bar{\Sigma}(p)$
Filtering input	estimated parameters	estimated parameters
Regime estimation	drawdown-threshold heuristic fit (Section 5.1)	statistical jump model (Section 5.2)
Critic steps per outer iteration	1	1
Multiplier EMA parameter (ω)	0.5	0.5
Stage 1 action bound	$a_{\max} = 1.0$, <code>cap_mode=component_tanh</code> (dollar-scale, componentwise tanh squash; Section 7.3 and Table 6)	same
Hard action projection	disabled (<code>apply_action_projection=false</code>); only the tanh squash above is applied	same
Actor learning rate α_ϕ	10^{-4}	same
Critic learning rate α_θ	3×10^{-4}	same
Multiplier learning rate α_ω	10^{-2}	same
Multiplier update frequency	every 10 outer iterations	same
Learning-rate schedule	step decay by 0.5 every 2000 iterations	same
Actor mixing weight η (Eq. (53))	0.3	same
Exploration covariance scale	0.1	same
Belief-dependent value-function term	enabled (<code>g_p_dep=true</code>)	same
Critic polynomial degree (m, m_g)	2, 2 (fixed, not separately tuned)	same
Gradient clipping norm	1.0	same
Advantage/signal normalization epsilon	10^{-8}	same
Checkpoint save frequency	every 250 iterations	same
Compute device	CPU	same

Table 6: Stage 2 (Center+DNNBand) configuration for the five-seed replication.

Item	Setting
Stage 1 center source	Common-covariance Stage 1 checkpoint, same training seed as the Stage 2 run
Filtering mode	Estimated parameters (<code>filter_mode = estimated_params</code>)
Transaction cost κ	0.002 (proportional, applied to pre-trade wealth)
Evaluation-time leverage cap	1.5, applied to the target weight at each rebalancing decision; not enforced during Stage 2 training and unrelated to the Stage 1 actor's action bound
Horizon / time step	1 year / 1/252 ($dt = 0.0039683$)
Stage 2 training iterations	800, with 128 simulated episodes per iteration
Training seeds	5 independent seeds (0–4)
Evaluation paths	1000 common held-out paths per training seed, evaluation seed 999
Evaluation scenario set	Same fixed set of 1000 held-out scenarios for all five training seeds (evaluation seed 999).
Policy architecture	Feedforward network, hidden width 64, separate softplus lower- and upper-gap heads
Optimizer / learning rate schedule	Adam, initial rate 5×10^{-4} , step decay by 0.5 every 100 iterations
Terminal objective	Log utility $U(x) = \log(\max(x, 10^{-10}))$ (<code>utility_gamma/gamma_risk</code> recorded but numerically inert for this configuration – see discussion below) plus turnover (5×10^{-4}), gap- L_2 (10^{-5}), and gross-leverage (10^{-5}) regularization, as in (113)
Checkpoint policy	Selected by validation utility every 25 iterations on 64 held-out validation paths.
Gross/net convention	Cumulative transaction cost is defined as gross terminal wealth minus net terminal wealth.

Tables 7 and 8 distinguish the true simulator from the estimated model used by filtering and control. This separation is essential: learning performance reflects both policy optimization and parameter-estimation error.

Table 7: True and estimated parameters: common-covariance benchmark.

Quantity	True environment	Estimated value used
$\mu^{(1)}$	(0.25, 0.18)	(0.3883, 0.2174)
$\mu^{(2)}$	(−0.73, −0.40)	(−0.4299, −0.3930)
Σ	$\begin{pmatrix} 0.0484 & 0.01188 \\ 0.01188 & 0.0324 \end{pmatrix}$	$\begin{pmatrix} 0.0490 & 0.0117 \\ 0.0117 & 0.0323 \end{pmatrix}$
λ_1	0.36	0.1641
λ_2	2.89	1.7500
Stationary regime-1 mass	0.889	0.914
Mean holding times	(2.78, 0.35) years	(6.09, 0.57) years

Table 8: True and SJM-estimated parameters: regime-dependent covariance run.

Quantity	True environment	SJM estimate used
$\mu^{(1)}$	(0.25, 0.18)	(0.3830, 0.2759)
$\mu^{(2)}$	(−0.73, −0.40)	(−0.3243, −0.1476)
$\Sigma^{(1)}$	$\begin{pmatrix} 0.0484 & 0.01188 \\ 0.01188 & 0.0324 \end{pmatrix}$	$\begin{pmatrix} 0.0485 & 0.0108 \\ 0.0108 & 0.0322 \end{pmatrix}$
$\Sigma^{(2)}$	$\begin{pmatrix} 0.0625 & 0.0225 \\ 0.0225 & 0.0484 \end{pmatrix}$	$\begin{pmatrix} 0.0574 & 0.0193 \\ 0.0193 & 0.0401 \end{pmatrix}$
λ_1	0.36	0.6801
λ_2	2.89	1.3310
Stationary regime-1 mass	0.889	0.662
Mean holding times	(2.78, 0.35) years	(1.47, 0.75) years

8.2 Repeated-seed robustness protocol

The common-covariance Stage 1 and Stage 2 experiments were repeated with five independent training seeds. Each trained policy was evaluated on the same fixed set of 1000 held-out market paths. Reported cross-seed

intervals are based on the five seed-level statistics and use the Student- t critical value with four degrees of freedom. These intervals quantify sensitivity to training initialization under the fixed evaluation design; they should not be interpreted as confidence intervals covering all sources of market-model or parameter uncertainty.

All policies were evaluated on the same deterministic set of 1000 held-out scenarios, generated from the fixed evaluation seed and path index. Paired differences were computed at the episode level, summarized within each training seed, and then aggregated across the five training seeds using the Student- t critical value with four degrees of freedom. Outputs from deterministic benchmark policies are identical across training seeds, providing an additional consistency check on scenario alignment.

The regime-dependent-covariance experiment reported in Section 8.3.1 was not part of this five-seed replication and remains a single-seed exploratory result (Section 8.3.1).

8.3 Stage 1 results

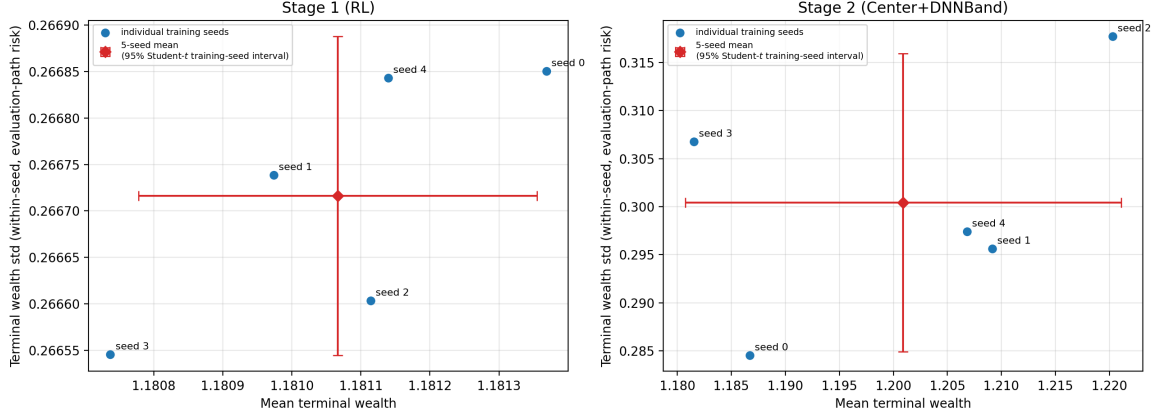
Table 9 summarizes the common-covariance benchmark across five independent training seeds, each evaluated on the same 1000 held-out paths. This case is closest to the theory because both the innovation filter and HJB reduction use common covariance.

Table 9: Common-covariance Stage 1 performance across five independent training seeds. Every entry is a 5-seed mean of the corresponding per-seed statistic (each per-seed statistic itself an average over that seed’s 1000 evaluation paths); bracketed values additionally report the 95% Student- t interval across the five training seeds (four degrees of freedom). Terminal-wealth dispersion (Std. X_T) is calculated across evaluation paths within each seed and is distinct from training-seed uncertainty. Equal weight and minimum variance do not depend on the trained policy and are seed-invariant by construction (zero-width interval).

Method	Mean X_T	Std. X_T (within seed)	$\mathbb{P}(X_T \geq z)$	Mean shortfall	Target MSD
Equal weight	1.0543 (seed-invariant)	0.2688	0.304	0.1943	0.0934
Minimum variance	1.0530 (seed-invariant)	0.2567	0.291	0.1913	0.0874
HJB-guided RL	1.1811 [1.1808, 1.1814]	0.2667	0.566	0.1247	0.0714

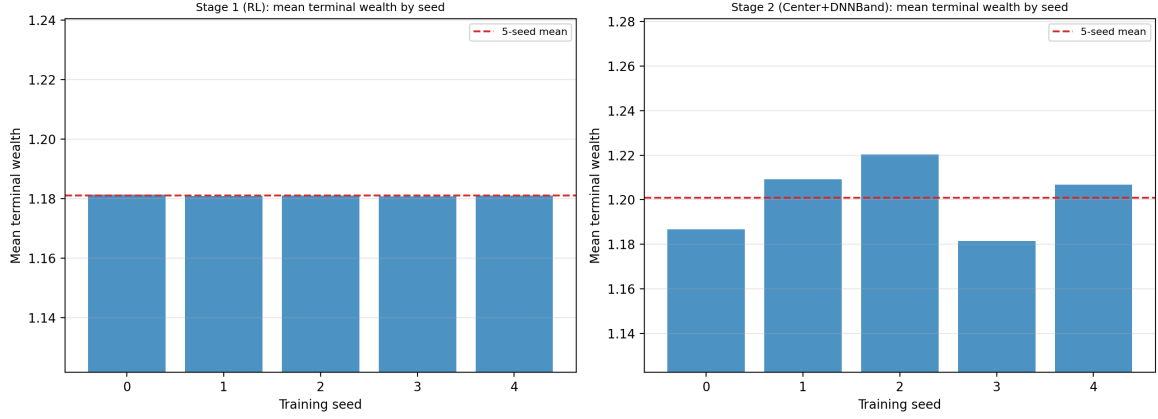
Across the five training seeds, HJB-guided RL achieved mean terminal wealth 1.1811 with a 95% training-seed interval of [1.1808, 1.1814], a target-achievement probability of 0.566, and a target mean squared deviation (target MSD) of 0.0714; target MSD is a derived metric computed as $\frac{1}{N_{\text{eval}}} \sum_e (X_{T,e} - z)^2$ from the evaluation terminal-wealth sample, not a native solver output. The between-training-seed standard deviation of mean terminal wealth was approximately 0.000232 – more than three orders of magnitude smaller than the within-seed evaluation-path dispersion (0.2667) – indicating that Stage 1 is extremely stable with respect to training initialization under this fixed evaluation design. Relative to minimum variance, HJB-guided RL raises mean terminal wealth by $1.1811 - 1.0530 = 0.1281$ and raises the target-achievement probability from 0.291 to 0.566; its within-seed standard deviation remains close to the static benchmarks, so the improvement is not explained solely by an increase in terminal dispersion.

Mean terminal wealth vs. terminal wealth dispersion, across 5 training seeds



(a) Mean terminal wealth vs. within-seed standard deviation, by training seed, with the five-seed mean and its 95% training-seed interval.

Training-seed variability - not a single representative run



(b) Mean terminal wealth by individual training seed, Stage 1 (left) and Stage 2 (right).

Figure 2: Training-seed robustness for the common-covariance HJB-guided policy (Stage 1, left panels of each row) and Center+DNNBand (Stage 2, right panels of each row). Both stages are shown together because the same five training seeds and evaluation design underlie both; Stage 1's near-total overlap of points contrasts with Stage 2's visibly larger spread.

8.3.1 Exploratory regime-dependent-covariance experiment

Table 10 reports a separate single-seed evaluation of the regime-dependent-covariance extension. Its purpose is to assess the implementability of the approximate filtering-and-control architecture; no multiple-seed inference is drawn from this experiment.

Table 10: Single-seed exploratory regime-dependent-covariance held-out simulation summary. Not part of the five-seed replication; no training-seed interval is reported.

Method	Mean X_T	Std. X_T	Var. X_T	$\mathbb{P}(X_T \geq z)$	Mean shortfall
Equal weight	1.078	0.271	0.073	0.316	0.180
Minimum variance	1.078	0.259	0.067	0.312	0.176
Approximate HJB RL	1.132	0.172	0.030	0.342	0.112

In this single exploratory run, the regime-dependent policy has the highest displayed mean and the lowest dispersion. Its gain in target-hit probability is more modest than in the common-covariance run. This is consistent with the SJM transition-rate mismatch and with the additional approximation in (71)–(76). The discrete-time Gaussian filtering-correction step is exact for the adopted observation model, but the belief-

averaged control layer is a moment-matched approximation with no verification theorem. These numbers demonstrate promising behavior for the implemented surrogate under a single training initialization; they do not prove optimality for the original regime-dependent continuous-time model, and no multi-seed inference or general dominance claim is made for this experiment.

8.4 Stage 2 results

Table 11 reports the five-seed transaction-cost-aware comparison.

Table 11: Stage 2 transaction-cost-aware performance across five independent training seeds, 1000 common evaluation paths per seed. Every entry is a 5-seed mean of the corresponding per-seed statistic (itself already an average over that seed’s 1000 evaluation paths, and additionally over the 252 within-path steps for Turnover and Band width); bracketed net-terminal-wealth values additionally report the 95% Student- t interval across the five training seeds. Gross X_T is an accounting decomposition (net wealth plus the transaction cost otherwise deducted, along the same realized trade path), not an independent counterfactual zero-cost re-simulation – see the metric-definition paragraph above and the discussion following this table. Cum. tc is defined as Gross X_T minus Net X_T . Band width is a structural quantity defined only for Center+DNNBand; “–” denotes not applicable, not zero.

Method	Net X_T	Gross X_T	Cum. tc	Turnover	Trades	Band width	$\mathbb{P}(X_T \geq z)$
Center+DNNBand	1.2009 [1.1807, 1.2211]	1.2196	0.0187	0.0397	175.8	1.740	0.490
CenterOnly, daily	1.1099 [1.1097, 1.1101]	1.1595	0.0496	0.1100	252.0	–	0.426
CenterOnly, monthly	1.1229 [1.1228, 1.1231]	1.1349	0.0120	0.0258	12.0	–	0.437
Equal weight, monthly	1.0372 (seed-invariant)	1.0401	0.0029	0.0059	12.0	–	0.281

Gross terminal wealth in Table 11 is not an independently re-simulated no-cost scenario: it shares the identical sequence of realized post-trade dollar positions and returns with net terminal wealth, and simply omits the transaction-cost debit at each step, so that by construction gross minus net equals cumulative transaction cost (verified to floating-point precision, Table 6). It does not represent what the policy would have traded, nor what it would have earned, in a world without transaction costs. Table 14 in Appendix C additionally reports 95% training-seed intervals for target achievement probability, target MSD, mean shortfall, cumulative transaction cost, turnover, trade count, band width, and mean terminal log-utility with its certainty equivalent (computed from episode-level terminal wealth using the same positivity floor as training, $\epsilon = 10^{-10}$) for every Stage 1 and Stage 2 method. Because only Stage 2 was trained against this log-utility objective, the Stage 2 values are objective-consistent evaluation metrics, whereas the Stage 1 values are post-hoc distributional summaries computed under an objective (log utility) different from the one Stage 1 was trained under (precommitted mean–variance).

Table 12: Selected Stage 2 paired comparisons. Differences are computed at the evaluation-path level within each training seed, averaged within the seed, and then aggregated across five training seeds; 95% intervals use the Student- t critical value with four degrees of freedom. The Stage 2-versus-Stage 1 row is a descriptive comparison across different terminal objectives, not a claim of dominance (Section 8.4). A complete set of eleven paired comparisons, including all pairwise method differences and gross-minus-net decompositions, is reported in Appendix C.

Comparison	Mean difference	95% training-seed interval	Excludes zero?
Center+DNNBand – CenterOnly, daily	0.0910	[0.0709, 0.1112]	yes
Center+DNNBand – CenterOnly, monthly	0.0780	[0.0578, 0.0981]	yes
Center+DNNBand – Equal weight, monthly	0.1637	[0.1436, 0.1839]	yes
Stage 2 net – Stage 1 terminal wealth	0.0198	[–0.0003, 0.0400]	no

Center+DNNBand achieved mean net terminal wealth 1.2009 (95% training-seed interval [1.1807, 1.2211]), mean gross terminal wealth 1.2196, mean cumulative transaction cost 0.0187, mean turnover 0.0397, a mean of 175.8 trades, and mean band width 1.740. Its paired difference against daily center rebalancing was 0.0910 with interval [0.0709, 0.1112], which excludes zero. The evaluated band-plus-cap Center+DNNBand policy reduced the mean number of trades from 252 under daily center rebalancing to 175.8, a reduction of approximately 30%; because the reported trade count is measured after the evaluation-time leverage cap of (107) is applied on top of the learned band, this reduction is attributable to the evaluated band-plus-cap policy rather than to the learned no-trade band alone. This reduction in trade frequency accompanies, but is not shown to causally

explain, the net-wealth improvement: separating the contribution of trade frequency from boundary asymmetry, belief conditioning, and time-to-go conditioning would require an ablation study that has not been performed. The mean descriptive difference between Stage 2 net terminal wealth and Stage 1 terminal wealth was 0.0198, with a 95% training-seed interval of $[-0.0003, 0.0400]$. Because the interval includes zero and the two stages optimize different terminal objectives (Stage 1 a precommitted mean-variance criterion, Stage 2 a transaction-cost-aware utility objective), this comparison does not establish that Stage 2 dominates Stage 1. It is also notable that, despite its higher mean terminal wealth, Center+DNNBand’s target-achievement probability (0.490) is lower than Stage 1’s (0.566): the two stages differ not only in mean level but in the shape of the terminal-wealth distribution relative to the target z , so policies should not be ranked by mean terminal wealth alone.

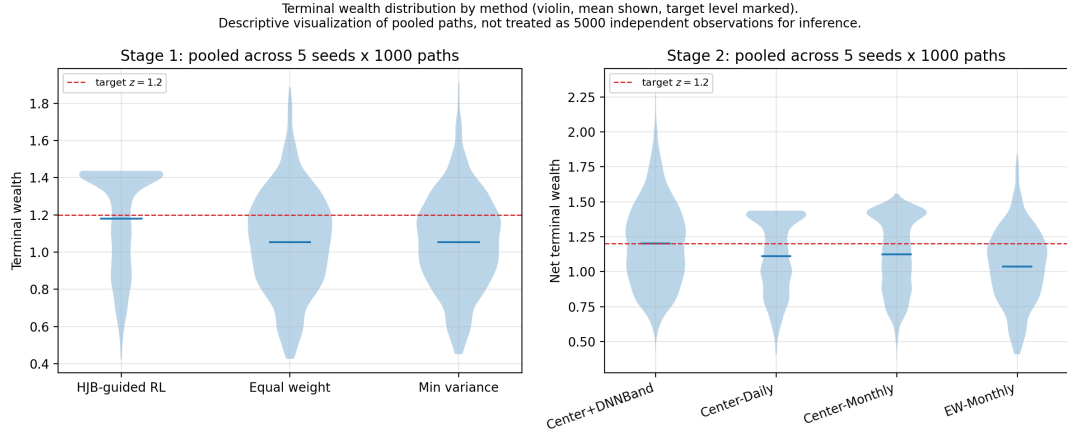


Figure 3: Terminal wealth distributions pooled across five training seeds and 1000 evaluation paths each. Left: Stage 1 (RL, equal weight, minimum variance). Right: Stage 2 (Center+DNNBand and the three comparators), net of transaction costs. Horizontal bars mark the pooled mean. These violin plots are descriptive visualizations of the pooled 5×1000 observations, not independent samples: the same 1000 evaluation scenarios recur under all five training seeds, so the pooled distribution mixes within-seed evaluation-path variability with across-seed policy variability rather than depicting 5000 independent draws.

Stage 2 cost and turnover comparison
(bars: 5-seed mean; error bars: 95% training-seed interval)

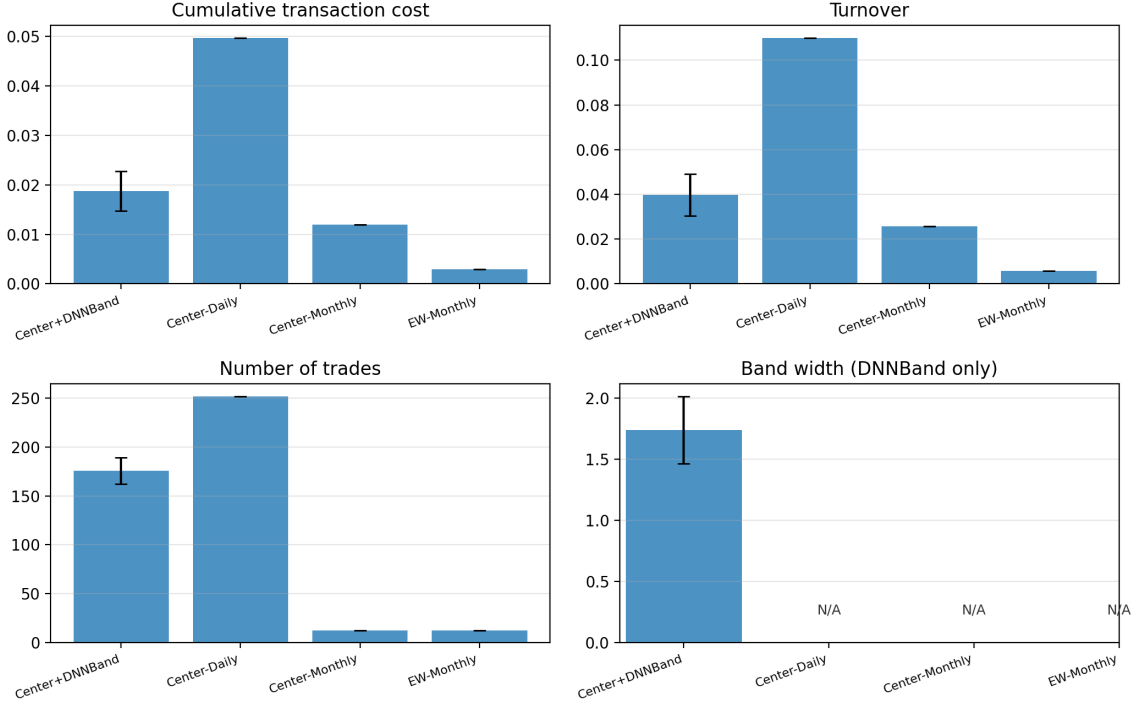


Figure 4: Stage 2 cost and turnover comparison across methods: cumulative transaction cost, turnover, trade count, and band width. Bars show the five-seed mean; error bars show the 95% training-seed interval. Band width is defined only for Center+DNNBand because the comparator policies do not contain a no-trade region. The “N/A” labels denote non-applicability, not zero-width bands.

9 Discussion

9.1 Practical implications

The results illustrate how an institutional investment rule can separate three economically distinct tasks: estimating the market environment, selecting a frictionless risk exposure, and deciding whether the benefit of rebalancing justifies its cost. The belief state is not merely a forecasting output; it enters the actor, critic, and transaction-cost policy as a decision variable. The band component provides an interpretable rebalancing rule: absent a leverage-cap violation, trading is deferred until the current position departs sufficiently from the belief- and horizon-dependent center. The evaluated implementation combines this band rule with the common leverage-feasibility projection described in Section 7. Consequently, a trade may be triggered either by a band crossing or by a leverage-cap violation.

This decomposition also clarifies why reinforcement learning is useful in this setting. The action is naturally a portfolio allocation, the state representation can be constructed to summarize economically relevant information, and the policy must account for the effect of current trading on future wealth and transaction opportunities. Incorporating domain knowledge into the state and policy does more than reduce dimension: the belief has a probabilistic interpretation, the Gaussian Stage 1 actor separates myopic and hedging demand, and the Stage 2 boundary explains when rebalancing is deferred. The same framework permits domain knowledge to enter the objective through target wealth, entropy, turnover, leverage, and transaction-cost terms.

Across the five-seed replication, the evaluated band-plus-cap policy reduced the mean number of trades by approximately 30% relative to daily center rebalancing (Section 8.4), which can be read as evidence that the framework offers an operational turnover-discipline benefit in addition to its net-performance statement, not only as a net-wealth result. These results are methodological demonstrations rather than investment recommendations: actual market impact, transaction-cost nonlinearity, and liability dynamics are not modeled here, and the reported experiments do not constitute a complete asset–liability management evaluation.

9.2 What is exact and what is approximate

The common-covariance Stage 1 model has the strongest theoretical status: filtering produces an observable diffusion, the entropy-regularized HJB yields the Gaussian optimizer, and the wealth-quadratic ansatz produces exact lower-dimensional coefficient PDEs under the stated regularity conditions. The SJM layer is an estimation methodology that supplies parameters to this filter; it is not part of the control verification theorem.

For regime-dependent covariance, the discrete-time Gaussian prediction–correction filter is exact for the specified hidden Markov observation model, but the belief-averaged effective diffusion and its HJB are moment-matched approximations. For transaction costs, the QVI is the theoretical benchmark and the center-relative critic/intervention actor is exploratory. Center+DNNBand, the reported Direct Boundary policy optimization method, produced a numerically usable pathwise policy across all five training seeds, with a separate terminal objective; it is not a numerical solution of that QVI.

9.3 Lessons from theory-guided learning

The experiments suggest an asymmetric role for structural theory. In Stage 1, the common-covariance HJB identifies the Gaussian policy family, the linear dependence of its mean on wealth, the quadratic critic, and the lower-dimensional coefficient functions. These restrictions remove large parts of the learning problem without materially narrowing the theorem-level optimizer. The resulting actor–critic provided usable training and favorable displayed results under both common and regime-dependent covariance. The very small between-seed dispersion in Stage 1 (between-training-seed standard deviation of mean terminal wealth ≈ 0.000232 , against a within-seed evaluation-path standard deviation of 0.267) is consistent with the stabilizing role of the HJB-restricted actor and critic, although this does not causally identify that restriction as the source of stability. Without a controlled structured-versus-unstructured ablation, the evidence does not identify the policy restriction itself as the sole cause of the observed stability.

The Stage 2 experience is different. The QVI identifies the enlarged state, the intervention-versus-continuation logic, and the relevance of a no-trade region, but it does not determine the finite-horizon multidimensional boundary geometry as sharply as the Stage 1 HJB determines the Gaussian actor. Jointly learning the QVI critic, obstacle residual, trigger, and intervention map was unstable. The Direct Boundary architecture instead imposed economically meaningful center-relative no-trade geometry, retained feasibility constraints and pathwise transaction costs, and produced a numerically usable implementation. Consistent with a less sharply identified boundary geometry, the between-training-seed standard deviation of mean net terminal wealth for the end-to-end Stage 1-plus-Stage 2 pipeline (each seed’s Stage 2 policy is trained against that same seed’s own Stage 1 center checkpoint, so the two stages’ initializations are not varied independently) was approximately 0.0163 – about 70 times larger than Stage 1 alone’s between-training-seed standard deviation of mean terminal wealth (≈ 0.000232) – indicating materially greater sensitivity to training initialization than Stage 1 under the same five-seed protocol. This reflects the combined pipeline’s sensitivity, not an isolated measurement of the Stage 2 optimizer with a fixed Stage 1 center held constant across seeds.

The methodological lesson is not that stronger structural restrictions are always preferable. Theory-guided parametrization is most effective when the underlying control equation identifies the relevant functional form with sufficient precision. When theory supplies only local or qualitative information about a free boundary, a softer inductive bias combined with simulation-based learning may be preferable to enforcing a misspecified low-dimensional geometry. The formula-based QVI actor remains valuable as an interpretable target and potential regularizer, but not yet as the reported optimizer.

9.4 Sources of performance and model risk

Performance in the regime-aware pipeline can be affected at five distinct layers:

- (1) **Regime identification error:** the SJM may assign observations to economically imperfect states.
- (2) **Parameter error:** regime-specific drifts, covariances, and transition intensities may be estimated inaccurately even conditional on the labels.
- (3) **Filtering error:** misspecified parameters distort the posterior belief supplied to the controller.
- (4) **Control approximation error:** the belief-averaged regime-dependent covariance model may differ from the original partially observed dynamics.
- (5) **Optimization error:** actor–critic or boundary training may converge slowly, become unstable, or settle at a poor local solution.

The exploratory regime-dependent experiment combines regime-identification, parameter-estimation, filtering, control-approximation, and optimization errors, so the reported performance difference cannot be attributed to any single layer. The SJM transition-rate mismatch provides direct evidence of parameter risk, while the additional f_p^2/f term and moment matching identify a separate control-model risk. The displayed approximate HJB policy has higher mean terminal wealth and lower dispersion than the static benchmarks, which supports the usefulness of the structural approximation in this setting, but it does not establish universal dominance or isolate the source of the improvement.

The SJM produced a practically usable persistent segmentation in the reported experiment. The analysis does not establish superiority over HMM-based estimation. A controlled comparison should examine label stability, parameter error, posterior filtering accuracy, and downstream portfolio performance under common data and evaluation paths. The evidence supports only the operational usefulness of combining persistent offline segmentation with online Bayesian belief updating.

Because the reported SJM features are constructed primarily from an equal-weight return index, covariance regimes are identified only indirectly through univariate aggregate-return features. Incorporating cross-sectional dispersion, realized correlation, or rolling covariance features is an important robustness extension, especially because covariance-state variation is a central motivation for the regime-dependent model.

The Stage 2 results have a similarly specific interpretation. Daily rebalancing to the frictionless center has lower mean net wealth than monthly rebalancing, consistent with repeated proportional costs offsetting the marginal tracking benefit. Center+DNNBand has the highest mean net wealth among the reported comparators and reduces mean trade count relative to daily rebalancing. Separate summaries of turnover, cumulative transaction cost, trade count, and band width provide a more direct view of implementation effects than net terminal wealth alone (Table 11, Figure 4). The available comparisons do not determine whether the net-wealth difference is driven primarily by trading frequency, boundary asymmetry, belief conditioning, or time-to-go conditioning; separating these mechanisms causally requires an ablation analysis that has not been performed.

9.5 Limitations and robustness status

The common-covariance Stage 1 and Stage 2 results are verified across five independent training seeds on a common evaluation design (Section 8.3–8.4). The following limitations remain:

- (1) Five training seeds are informative about sensitivity to training initialization but remain a small sample; the reported Student- t intervals should not be read as tight population-level confidence statements.
- (2) The regime-dependent-covariance experiment (Section 8.3.1) remains single-seed and exploratory; it is not part of the five-seed replication.
- (3) Transaction-cost sensitivity (varying κ and the leverage cap) has not been examined.
- (4) An actor-signal ablation separating the martingale and full-horizon terms, and a structured-versus-unstructured policy-class ablation, have not been performed.
- (5) The evaluation output contains terminal wealth and path-level summary statistics but not complete time-series wealth trajectories, precluding a time-resolved multi-seed wealth-path analysis.
- (6) A time-resolved leverage-cap saturation measure is not available.
- (7) Only the combined band width is available; separate lower- and upper-boundary summaries are not reported.
- (8) Stage 1 and Stage 2 optimize different terminal objectives, so comparisons between them (Section 8.4) are descriptive, not a test of dominance under a common objective.
- (9) Filtering quality depends on the regime segmentation and parameter estimates, and the estimated transition persistence differs materially from the simulator in the regime-dependent-covariance experiment; the associated control layer is moment-matched rather than exact, and $\tilde{c}(p)$ has not been identified from a continuous-time filtering theorem.
- (10) The attempted QVI-based RL was unstable, and hard componentwise projection in Center+DNNBand is interpretable but does not represent the most general multidimensional intervention geometry.
- (11) Liabilities, solvency capital, taxes, market impact, and governance constraints are not explicitly modeled.

9.6 Future research

The transaction-cost QVI derived in Section 7 provides the theoretical target,

$$\min\{-V_t - \mathcal{L}V, \mathcal{M}V - V\} = 0, \quad (118)$$

where $\mathcal{M}$ accounts for proportional wealth loss and a post-trade target map. The next methodological step is not to formulate this QVI again, but to stabilize its numerical learning: continuation/intervention sampling, obstacle-compatible losses, target networks for $\mathcal{M}$, curriculum learning from a Direct Boundary policy, and smooth trigger approximations are natural candidates.

Broader sensitivity to transaction-cost rates, entropy weights, and filtering misspecification, together with ablations separating the effects of the belief state, HJB-guided parametrization, martingale signal, full-horizon residual, symmetric versus asymmetric bands, and QVI-derived priors, remain directions for future work. Stronger benchmarks should include a full-information regime oracle retrained under that information structure, a true-parameter filtered policy, a myopic policy without the belief-hedging term, and a fixed symmetric band. Supplying the true regime directly to a policy trained only on filtered beliefs would instead be an out-of-distribution stress test, not a formal oracle comparison. An actuarial extension should introduce stochastic liability cash flows, funding-ratio or surplus objectives, regulatory capital constraints, and asset eligibility limits so that the portfolio controller becomes a component of a complete ALM problem.

10 Conclusion

This paper starts from the view that reinforcement learning is well matched to dynamic portfolio choice when the economically relevant state and admissible policy class are constructed carefully from control theory rather than left unrestricted. The central contribution is a belief-state HJB reduction of partially observed mean–variance control, extended from common to regime-dependent observation covariance; building on this reduction, Stage 1 makes a structure-guided actor–critic operational, and Stage 2 learns a transaction-cost no-trade boundary around the frozen Stage 1 center through Direct Boundary policy optimization. The resulting architecture combines state estimation, sequential control, and rebalancing discipline without treating portfolio selection as an unrestricted state-to-weight prediction problem.

The five-seed common-covariance study gives different robustness conclusions for the two learning stages. Stage 1 converged to nearly identical held-out performance across training seeds, with mean terminal wealth 1.1811 and very small between-seed dispersion. Center+DNNBand achieved the highest average net terminal wealth among the reported transaction-cost-aware comparators, retained a positive paired difference relative to daily and monthly center rebalancing, and reduced mean trade count by approximately 30% relative to daily rebalancing under the evaluated band-plus-cap policy; however, it was materially more sensitive to training initialization, and its descriptive net-wealth difference from Stage 1 had a 95% interval that included zero. The evidence therefore supports robustness relative to the reported Stage 2 comparators, not a claim that Stage 2 statistically dominates the frictionless Stage 1 policy. The regime-dependent-covariance extension remains exploratory and single-seed (Section 8.3.1) and is not covered by these five-seed robustness statements. Center+DNNBand is not a numerical solution of the transaction-cost QVI derived in this paper; it is a structurally motivated but separately optimized implementation.

The broader methodological conclusion is that the contribution lies not in deep learning alone, but in selectively using financial structure to determine the information state, restrict policy geometry, and design the objective, while retaining flexible learning where the control theory does not identify the optimal functional form sufficiently sharply. Stage 1 uses strong low-dimensional restrictions because the Gaussian actor and quadratic critic are characterized by the HJB; in Stage 2, the free-boundary geometry is less completely identified, so flexible simulation-based boundary learning provided the usable implementation, while the QVI and center-relative formulas remain theoretical and architectural guides. For actuarial and institutional applications – where the target wealth can serve as a funding or accumulation objective and the no-trade band as an operational turnover-discipline rule – this combination of interpretable belief states, structured policies, and transaction-aware rebalancing offers a tractable portfolio-control building block, not a complete asset–liability management solution; a full ALM extension would additionally require stochastic liabilities, capital constraints, and institution-specific admissibility rules (Section 1).

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A Explicit QVI Continuation Coefficients

For compactness, suppress the arguments $(p, \mathbf{w})$ and define

$$s = \|\bar{\sigma}\|^2, \quad q = \|\beta\|^2, \quad R = DD^\top, \quad c_p = \bar{\sigma}^\top \beta, \quad c_w = D\bar{\sigma}, \quad c_{pw} = D\beta. \quad (119)$$

The generator associated with (81) is

$$\begin{aligned} \mathcal{L}V = & x\bar{\mu}V_x + \mu_p V_p + b_w^\top \nabla_w V + \frac{1}{2}x^2 s V_{xx} + \frac{1}{2}q V_{pp} + \frac{1}{2} \text{tr}(R \nabla_{ww}^2 V) \\ & + x c_p V_{xp} + x c_w^\top \nabla_w V_x + c_{pw}^\top \nabla_w V_p. \end{aligned} \quad (120)$$

Substitution of (84) into (86) gives

$$\begin{aligned} \Phi_2 = & -A_t - 2\bar{\mu}A - sA - \mu_p A_p - b_w^\top \nabla_w A - \frac{1}{2}q A_{pp} - \frac{1}{2} \text{tr}(R \nabla_{ww}^2 A) \\ & - 2c_p A_p - 2c_w^\top \nabla_w A - c_{pw}^\top \nabla_w A_p, \end{aligned} \quad (121)$$

$$\begin{aligned} \Phi_1 = & -B_t - \bar{\mu}(B + 2\omega A) - 2\omega s A - \mu_p B_p - b_w^\top \nabla_w B - \frac{1}{2}q B_{pp} - \frac{1}{2} \text{tr}(R \nabla_{ww}^2 B) \\ & - c_p(B_p + 2\omega A_p) - c_w^\top (\nabla_w B + 2\omega \nabla_w A) - c_{pw}^\top \nabla_w B_p, \end{aligned} \quad (122)$$

$$\begin{aligned} \Phi_0 = & -C_t - \omega \bar{\mu}B - \omega^2 s A - \mu_p C_p - b_w^\top \nabla_w C - \frac{1}{2}q C_{pp} - \frac{1}{2} \text{tr}(R \nabla_{ww}^2 C) \\ & - \omega c_p B_p - \omega c_w^\top \nabla_w B - c_{pw}^\top \nabla_w C_p. \end{aligned} \quad (123)$$

Equations (121)–(123) show why a third coefficient B is required: even if $B(T, p, \mathbf{w}) = 0$, the terms proportional to ωA act as sources for the linear equation away from maturity.

B Center-Relative QVI Critic and Intervention Actor

Let $\tau = T - t$, $\delta = \mathbf{w} - \mathbf{w}^{\text{ref}}(t, p)$, and

$$\|\delta\|_{M(p)}^2 = \delta^\top M(p) \delta, \quad (124)$$

where $M(p)$ may be I , $\bar{\Sigma}(p)^{-1}$, or a diagonal inverse-variance metric. A low-dimensional critic consistent with positivity of A and the parity of the coefficient equations is

$$A_\theta(\tau, p, \delta) = \exp \left(a_0(\tau, p) + a_2(\tau, p) \|\delta\|_{M(p)}^2 + a_4(\tau, p) \|\delta\|_{M(p)}^4 \right), \quad (125)$$

$$B_\theta(\tau, p, \delta) = b_1(\tau, p)^\top \delta + b_3(\tau, p)^\top \delta \|\delta\|_{M(p)}^2, \quad (126)$$

$$C_\theta(\tau, p, \delta) = c_0(\tau, p) + c_2(\tau, p) \|\delta\|_{M(p)}^2 + c_4(\tau, p) \|\delta\|_{M(p)}^4. \quad (127)$$

Each coefficient surface in (τ, p) can be represented by a low-order polynomial, as in Stage 1. The intervention actor may be written

$$\Gamma_\phi(t, p, \mathbf{w}) = \mathbf{w} + G_\phi(\tau, p, \delta) = \mathbf{w}^{\text{ref}}(t, p) + \delta + G_\phi(\tau, p, \delta), \quad (128)$$

with a radial-directional model

$$G_\phi = -s_\phi(\tau, p, \delta)\delta + R_\phi(\tau, p, \delta), \quad s_\phi = \text{sigmoid} \left(\eta_0(\tau, p) + \eta_2(\tau, p) \|\delta\|_{M(p)}^2 + \eta_4(\tau, p) \|\delta\|_{M(p)}^4 \right). \quad (129)$$

The radial first model sets $R_\phi \equiv 0$, so $s_\phi = 0$ gives no trade and $s_\phi = 1$ moves directly to the normalized reference center. This ensures $G_\phi(\tau, p, 0) = 0$ and biases interventions toward that center. A feasibility layer can enforce simplex, box, or leverage constraints. Because the implemented center (102) depends on wealth, this appendix basis is an architectural approximation and does not claim exact closure after the wealth-dependent change of coordinates.

For the cost-minimization convention, define

$$R_{\text{cont}} = -V_{\theta,t} - \mathcal{L}V_\theta, \quad R_{\text{obs}} = \mathcal{M}_\phi V_\theta - V_\theta. \quad (130)$$

The exact QVI requires

$$R_{\text{cont}} \geq 0, \quad R_{\text{obs}} \geq 0, \quad R_{\text{cont}} R_{\text{obs}} = 0. \quad (131)$$

A soft residual objective may combine terminal supervision with penalties for negative continuation/obstacle residuals and complementarity:

$$\mathcal{L}_{\text{QVI}} = \mathcal{L}_{\text{terminal}} + \lambda_c \mathbb{E}[(R_{\text{cont}}^-)^2] + \lambda_o \mathbb{E}[(R_{\text{obs}}^-)^2] + \lambda_{co} \mathbb{E}[(R_{\text{cont}} R_{\text{obs}})^2], \quad (132)$$

where $r^- = \min(r, 0)$. This residual objective defines the QVI-based alternative considered in the study. Under the tested optimization scheme, simultaneous fitting of the residuals and intervention policy was unstable; the reported Stage 2 results therefore use Direct Boundary policy optimization.

C Complete Paired Comparisons and Metric Intervals

Table 13 reports all eleven Stage 2 paired comparisons computed for the five-seed replication, extending the four comparisons selected for the main text (Table 12). Methodology is identical: differences are computed at the evaluation-path level within each of the five training seeds, averaged within the seed, and then aggregated across seeds using the Student- t critical value with four degrees of freedom. Rows marked “descriptive” compare methods with different rebalancing rules or, in the Stage 2-versus-Stage 1 case, different terminal objectives; they are not dominance claims.

Table 13: Complete set of paired comparisons, 5 training seeds $\times$ 1000 evaluation paths per seed.

Comparison	Mean diff.	95% interval	Excl. zero?	Type
Stage 2 net – Stage 1 terminal wealth	0.0198	[−0.0003, 0.0400]	no	descriptive
Gross – net, Center+DNNBand	0.0187	[0.0148, 0.0227]	yes	cost decomposition
Gross – net, CenterOnly daily	0.0496	[0.0496, 0.0497]	yes	cost decomposition
Gross – net, CenterOnly monthly	0.0120	[0.0120, 0.0120]	yes	cost decomposition
Gross – net, EW monthly	0.0029	[0.0029, 0.0029]	yes	cost decomposition
Center+DNNBand – CenterOnly daily	0.0910	[0.0709, 0.1112]	yes	descriptive
Center+DNNBand – CenterOnly monthly	0.0780	[0.0578, 0.0981]	yes	descriptive
Center+DNNBand – EW monthly	0.1637	[0.1436, 0.1839]	yes	descriptive
CenterOnly daily – CenterOnly monthly	−0.0131	[−0.0131, −0.0130]	yes	descriptive
CenterOnly daily – EW monthly	0.0727	[0.0725, 0.0729]	yes	descriptive
CenterOnly monthly – EW monthly	0.0858	[0.0856, 0.0860]	yes	descriptive

Table 14 reports 95% Student- t training-seed intervals for evaluation metrics not already fully tabulated with intervals in the main text (Tables 9 and 11 report only the terminal-wealth interval). Mean terminal log-utility and its certainty equivalent are reported for the HJB-guided Stage 1 policy and all four Stage 2 comparators, but not for the static Stage 1 equal-weight and minimum-variance baselines. Only the Stage 2 values are objective-consistent, since Stage 2 is trained directly against this log-utility objective; the Stage 1 value is a post-hoc distributional summary evaluated under an objective different from the precommitted mean–variance criterion Stage 1 was actually trained under, and should not be read as evidence about Stage 1’s performance under its own objective. “–” denotes a metric not applicable to that method (e.g. band width for non-DNN-band Stage 2 comparators, or Stage 2-specific cost/turnover metrics for the Stage 1 policies, which do not trade with proportional cost in the Stage 1 evaluation).

Table 14: Distributional metrics and post-hoc log-utility summaries, 5-seed mean with 95% Student- t interval across training seeds (Stage 1: within-seed statistic over 1000 paths; Stage 2: as Table 11). Target MSD and mean shortfall are derived from evaluation terminal wealth, not native solver outputs. Mean terminal utility and its log-utility certainty equivalent are computed from episode-level terminal wealth using $U(x) = \log(\max(x, 10^{-10}))$, the exact positivity floor used in training; these are objective-consistent only for Stage 2, which was trained against this objective, and are post-hoc distributional summaries (not objective-consistent) for the Stage 1 row, which was trained under a precommitted mean–variance criterion.

Stage	Method	$\mathbb{P}(X_T \geq z)$	Target MSD	Mean shortfall	Mean terminal utility (cert. equiv.)
1	HJB-guided RL	0.5664 [0.5657, 0.5671]	0.0714 [0.0713, 0.0715]	0.1247 [0.1246, 0.1248]	0.1358 [0.1356, 0.1361] (1.1455)
1	Equal weight	0.3040 (seed-invariant)	0.0934 (seed-invariant)	0.1943 (seed-invariant)	–
1	Minimum variance	0.2910 (seed-invariant)	0.0874 (seed-invariant)	0.1913 (seed-invariant)	–
2	Center+DNNBand	0.4902 [0.4699, 0.5105]	0.0905 [0.0810, 0.1000]	0.1229 [0.1126, 0.1332]	0.1509 [0.1338, 0.1679] (1.1629)
2	CenterOnly, daily	0.4260 [0.4242, 0.4278]	0.0780 [0.0780, 0.0781]	0.1633 [0.1632, 0.1633]	0.0722 [0.0720, 0.0724] (1.0749)
2	CenterOnly, monthly	0.4368 [0.4362, 0.4374]	0.0739 [0.0738, 0.0739]	0.1538 [0.1538, 0.1539]	0.0859 [0.0858, 0.0861] (1.0897)
2	Equal weight, monthly	0.2810 (seed-invariant)	0.0970 (seed-invariant)	0.2049 (seed-invariant)	0.0001 (seed-invariant) (1.0001)

Table 15: Stage 2 cost/turnover metrics, 5-seed mean with 95% Student- t interval across training seeds; each per-seed value is already averaged over 1000 evaluation paths and, for turnover and band width, over the 252 within-path steps (see the metric-definition paragraph, Section 8.1).

Method	Cumulative tc	Turnover	Trade count	Band width
Center+DNNBand	0.0187 [0.0148, 0.0227]	0.0397 [0.0303, 0.0491]	175.8 [162.0, 189.5]	1.740 [1.465, 2.015]
CenterOnly, daily	0.0496 [0.0496, 0.0497]	0.1100 [0.1099, 0.1100]	252.0 (seed-invariant)	–
CenterOnly, monthly	0.0120 (seed-invariant)	0.0258 (seed-invariant)	12.0 (seed-invariant)	–
Equal weight, monthly	0.0029 (seed-invariant)	0.0059 (seed-invariant)	12.0 (seed-invariant)	–

D Code and Reproducibility Materials

The implementation, experiment configurations, and scripts used to produce the tables and figures are available from the author upon reasonable request. All numerical experiments use synthetic data generated from the models specified in the paper. Reproducibility of the exploratory regime-dependent-covariance experiment is limited by the incomplete specification described in Section 5.2.

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